

**Comparative genomics and evolutionary analysis of the
ERF gene family in sugarcane and its wild relatives**

**Project report submitted in partial fulfilment of the
requirements for the award of Master of Science**

by

**KEERTHANA P K
24MBI016**



Under the Guidance of

**Dr. C. Nilavamuthan
Assistant Professor**

**Department of Bioscience
Sri Krishna Arts and Science College
Affiliated to Bharathiar University
Coimbatore - 641008.**

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CERTIFICATE

This is to certify that the Project report entitled ***“Comparative genomics and evolutionary analysis of the ERF gene family in Sugarcane and its wild relatives”*** in partial fulfilment of the requirements for the award of the degree of Master of Science to Sri Krishna Arts and Science College, affiliated to Bharathiar University, Coimbatore, is a record of bonafide work carried out by **KEERTHANA P K (24MBI016)** and that no part of this has been submitted for the award of any other degree or diploma and the work has not been published in popular journal or magazine.

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Affiliated to Bharathiar University
Kuniamuthur, Coimbatore - 641008

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I hereby declare that the Project report entitled ***“Comparative genomics and evolutionary analysis of the ERF gene family in sugarcane and its wild relatives”*** submitted to Sri Krishna Arts and Science College, affiliated to Bharathiar University, Coimbatore, in partial fulfilment of the requirements for the award of the degree of Master of Science is an original work and it has not been previously formed the basis for the award of any Degree, Diploma, Associateship, Fellowship or similar titles to any other university or body during the period of my study.

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ABSTRACT

The APETALA2/Ethylene Responsive Factor (AP2/ERF) superfamily represents one of the most critical regulatory networks in plants, mediating responses to environmental stressors and developmental programs. In the highly complex polyploid sugarcane genome, understanding the organization of this superfamily is essential for enhancing climate resilience and sugar yield. This study employed a comprehensive bioinformatic and comparative genomic pipeline to identify and characterize the AP2/ERF superfamily across three distinct *Saccharum* lineages: the noble *S. officinarum*, the wild ancestor *Saccharum spontaneum*, and the commercial hybrid R570.

The results revealed a dramatic 7.5-fold genomic proliferation during the "nobilization" process, with the AP2/ERF family expanding from 176 genes in *S. spontaneum* to 1,336 genes in *S. officinarum*. Phylogenetic reconstruction partitioned these expansive datasets into established subgroups (I–X), identifying significant "Lineage-Specific Expansions" (LSE) in Groups VII, III, and IX, which are associated with flooding, drought, and biotic defense, respectively. Structural analysis showed a predominantly intronless architecture across all lineages, supporting the "Speedy Transcription" theory for rapid environmental response. Furthermore, chromosomal mapping and synteny analysis using MCScanX confirmed that this expansion was driven by Segmental and Whole-Genome Duplications (WGD), providing a robust "genetic safety net" or dosage-buffering effect.

This research establishes a definitive molecular baseline for the sugarcane AP2/ERF regulatory network. The high level of syntenic retention observed in the R570 hybrid demonstrates that the core stress-response machinery of the noble ancestor is successfully inherited during breeding. Ultimately, this work provides a high-resolution genetic framework for the application of Marker-Assisted Selection (MAS) and CRISPR/Cas9 genome editing, transitioning sugarcane improvement from reactive phenotypic selection to a proactive, genomic-based strategy for long-term agricultural sustainability.

Keywords: *Saccharum* complex, AP2/ERF Superfamily, Polyploidy, Genome-wide Identification, Phylogenetic Analysis, Synteny, Abiotic Stress, Nobilization.

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1. INTRODUCTION

Sugarcane (*Saccharum* spp.) is one of the most economically and industrially important crops cultivated worldwide, serving as the primary source of sugar and a major feedstock for bioethanol production. It contributes nearly 75–80% of global sugar production and plays a crucial role in renewable energy generation, particularly in countries such as India and Brazil. In India, sugarcane cultivation supports a vast agro-based industry and provides livelihood to millions of farmers. In addition to sugar and ethanol, sugarcane is also utilized in the production of paper, biofertilizers, animal feed, and various by-products, making it a multipurpose crop of high economic value. Despite its importance, the genetic improvement of sugarcane remains a significant challenge due to its highly complex genome structure and limited genetic variability available for breeding programs. (Meena et al., 2024)

Sugarcane possesses one of the most complex genomes among cultivated crops. It is a highly polyploid and aneuploid species, derived from interspecific hybridization between *Saccharum officinarum* (*S. officinarum* (high sugar content)) and *Saccharum Spontaneum* (*S. spontaneum*(stress tolerance and adaptability)). Modern sugarcane cultivars typically exhibit chromosome numbers ranging from $2n = 100$ to 130 and contain a mixture of homologous and homeologous chromosomes. The genome is characterized by extensive heterozygosity, high levels of gene duplication, and the presence of repetitive DNA sequences. These features result in functional redundancy among genes, making it difficult to assign specific functions to individual gene copies. Moreover, uneven chromosome pairing during meiosis complicates genetic inheritance and breeding efforts. The large genome size and polyploid nature also pose challenges for genome sequencing, assembly, and annotation. Consequently, advanced bioinformatics and comparative genomics approaches are essential for identifying and characterizing gene families in sugarcane. (Ali et al., 2019)

Plants are constantly exposed to a wide range of environmental stresses that adversely affect their growth, development, and productivity. These stresses include both biotic factors, such as pathogens and pests, and abiotic factors, including drought, salinity, extreme temperatures, and waterlogging. Among these, abiotic stresses are major limiting factors for crop yield worldwide. In sugarcane, drought stress leads to reduced photosynthesis, impaired growth, and decreased sucrose accumulation, while waterlogging results in oxygen deficiency in the root zone, leading to hypoxic or anoxic conditions. Under such conditions, cellular respiration is severely affected, resulting in reduced ATP production and accumulation of toxic metabolites.

To survive under these adverse conditions, plants have evolved complex regulatory networks that involve stress perception, signal transduction, transcriptional regulation, and activation of stress-responsive genes. (Mustafa et al., 2018)

Transcription factors play a central role in these regulatory networks by controlling gene expression in response to environmental stimuli. Among the various transcription factor families in plants, the AP2/ERF (APETALA2/Ethylene Responsive Factor) superfamily is one of the largest and most important groups. Members of this family are unique to plants and are characterized by the presence of a conserved AP2 DNA-binding domain consisting of approximately 60–70 amino acids. Structurally, AP2/ERF transcription factors contain several functional regions, including the DNA-binding domain, transcriptional activation or repression domain, oligomerization domain, and nuclear localization signal. The AP2 domain contains conserved YRG and RAYD motifs, which are essential for DNA binding and protein stability. These transcription factors regulate gene expression by binding to specific cis-regulatory elements such as the GCC-box and DRE/CRT elements present in the promoters of target genes. (Liu et al., 2022)

The AP2/ERF superfamily is classified into five major groups based on domain organization and sequence similarity: AP2, ERF, DREB, RAV, and Soloist. The AP2 group typically contains two AP2 domains and is mainly involved in developmental processes such as floral organ identity and seed development. The ERF and DREB groups contain a single AP2 domain but differ in their DNA-binding specificity and functional roles. ERF proteins bind to the GCC-box and are primarily associated with ethylene signaling and biotic stress responses, whereas DREB proteins bind to the DRE/CRT element and are involved in abiotic stress responses such as drought, salinity, and cold stress. The RAV group contains both AP2 and B3 domains and is involved in hormone signaling pathways, while the Soloist group represents a unique and less-characterized lineage within the superfamily. (Abiri et al., 2017)

Further phylogenetic classification has divided the ERF and DREB families into multiple subgroups based on conserved motifs and sequence similarity. These subgroups, often designated as Groups I to X (and sometimes up to XII depending on classification systems), reflect the evolutionary diversification of the ERF gene family. For example, Groups I–IV are generally associated with DREB proteins involved in abiotic stress responses, while Groups V–X correspond to ERF proteins involved in biotic stress and hormonal regulation. Each subgroup exhibits distinct conserved motifs beyond the AP2 domain, which contribute to their

specific functional roles. Gene duplication events, including tandem and segmental duplications, have played a significant role in the expansion of these subgroups, leading to functional specialization and adaptation to diverse environmental conditions. (Sakuma et al., 2002)

AP2/ERF transcription factors are involved in a wide range of biological processes beyond stress responses. They play important roles in plant growth and development, including regulation of cell division, organ formation, and senescence. They are also involved in hormonal signaling pathways such as ethylene, abscisic acid (ABA), jasmonic acid (JA), and gibberellins (GA), which regulate plant responses to environmental stimuli. For example, ERF proteins are key regulators of ethylene signaling pathways, which are involved in fruit ripening, leaf senescence, and stress responses. In addition, AP2/ERF genes have been implicated in regulating secondary metabolism, including the biosynthesis of flavonoids, alkaloids, and other important compounds. (Nakano et al., 2006)

A particularly important subgroup within the ERF family is the ERF-VII group, which plays a central role in plant adaptation to hypoxic conditions caused by waterlogging and flooding. ERF-VII transcription factors are regulated through the N-end rule pathway, which controls protein stability in response to oxygen and nitric oxide levels. Under normal oxygen conditions, ERF-VII proteins are targeted for degradation, whereas under hypoxic conditions, they are stabilized and accumulate in the nucleus, where they activate the expression of hypoxia-responsive genes. These genes are involved in anaerobic metabolism, including glycolysis and fermentation pathways, which enable plants to generate energy under low oxygen conditions. (Xu et al., 2006)

The role of ERF-VII transcription factors has been extensively studied in *Oryza sativa*, where the SUB1A gene confers submergence tolerance by suppressing elongation growth and conserving energy. In contrast, SNORKEL genes promote rapid elongation growth, allowing plants to escape flooding conditions. In *Arabidopsis thaliana*, ERF-VII genes such as RAP2.2, RAP2.3, RAP2.12, HRE1, and HRE2 regulate hypoxia-responsive gene expression and play key roles in oxygen sensing and signaling. These transcription factors are known to interact with ethylene signaling pathways, further highlighting the complex regulatory networks involved in stress responses. (Hattori et al., 2009)

Genome-wide studies in various crops have revealed the widespread distribution and functional importance of AP2/ERF genes. In *Arabidopsis thaliana* and *Oryza sativa*, the number of ERF genes

reported varies across studies depending on genome annotation and classification criteria, with estimates ranging from approximately 120–140 in *Arabidopsis* and 130–160 in *Oryza sativa*. In wheat, maize, soybean, and other crops, AP2/ERF genes have been shown to play critical roles in stress tolerance, growth regulation, and development. In sugarcane, particularly in *Saccharum spontaneum*, a large number of AP2/ERF genes have been identified, although the exact number varies among studies due to differences in analysis pipelines and filtering approaches. These variations highlight the complexity of gene family identification and the influence of methodological differences. (Hinz et al., 2010)

In addition to waterlogging, ERF genes are also involved in responses to other abiotic stresses such as drought, salinity, and temperature extremes. DREB proteins, for example, regulate genes involved in osmotic stress tolerance, while ERF proteins are involved in pathogen defense and biotic stress responses. The ability of AP2/ERF transcription factors to regulate multiple stress pathways makes them key targets for crop improvement programs. (Gasch et al., 2016)

The expansion and diversification of the AP2/ERF gene family in plants are largely attributed to gene duplication events and polyploidization, which are key drivers of genome evolution. Gene duplication provides raw genetic material for evolutionary innovation, allowing duplicated genes to acquire new functions (neofunctionalization), partition existing functions (subfunctionalization), or become non-functional through pseudogenization. In plant genomes, duplication events are commonly categorized into tandem duplication, segmental duplication, and whole-genome duplication (polyploidy), all of which have contributed significantly to the expansion of transcription factor families, including AP2/ERF. (Li et al., 2020)

Among these mechanisms, whole-genome duplication plays a particularly important role in crops such as sugarcane, which has undergone multiple rounds of polyploidization during its evolutionary history. Modern sugarcane cultivars are derived from interspecific hybridization between *S. officinarum* and *S. spontaneum*, followed by repeated backcrossing and chromosome doubling. This has resulted in a highly polyploid and aneuploid genome with extensive gene redundancy. Such genomic complexity leads to the presence of multiple copies of ERF genes, often distributed across different homologous chromosomes. These duplicated genes may retain similar sequences but can exhibit differences in expression patterns, regulatory elements, and functional roles. (Meena et al., 2024)

Tandem duplication, which involves the duplication of genes in close proximity on the same chromosome, is another important mechanism contributing to the expansion of the ERF gene family. Tandemly duplicated ERF genes often form gene clusters and may be co-regulated or functionally related. These clusters can rapidly evolve, allowing plants to adapt to changing environmental conditions. Segmental duplication, on the other hand, involves duplication of larger chromosomal regions and contributes to the dispersal of ERF genes across the genome. Both tandem and segmental duplications have been reported in several plant species, including *Arabidopsis thaliana*, *Oryza sativa*, and maize, highlighting their role in shaping the structure of the ERF gene family. (Ali et al., 2019)

In polyploid species like sugarcane, the retention of duplicated genes is often influenced by selective pressures. Genes involved in stress responses, such as ERF transcription factors, are more likely to be retained due to their adaptive significance. This retention allows plants to fine-tune their responses to environmental stresses through differential gene expression. For instance, duplicated ERF genes may be expressed in different tissues, developmental stages, or stress conditions, thereby increasing the regulatory flexibility of the plant. This phenomenon is particularly important in sugarcane, which is frequently exposed to multiple abiotic stresses such as drought, salinity, and waterlogging. (Mustafa et al., 2018)

Evolutionary analysis of ERF genes across different plant species has revealed both conservation and divergence within the gene family. The conserved AP2 domain indicates a common evolutionary origin, while variations in other regions of the protein contribute to functional diversification. Comparative genomics studies have shown that many sugarcane ERF genes share high sequence similarity with those in closely related species such as sorghum, reflecting their shared ancestry. However, species-specific expansions and unique gene copies are also observed, suggesting adaptation to distinct environmental conditions and ecological niches. (Liu et al., 2022)

Furthermore, regulatory divergence plays a crucial role in the functional evolution of duplicated ERF genes. Changes in promoter regions, cis-regulatory elements, and transcription factor binding sites can lead to differences in gene expression patterns, even among highly similar gene copies. Epigenetic modifications such as DNA methylation and histone modifications may also influence gene expression, contributing to the complexity of ERF gene regulation in polyploid genomes. These regulatory variations enable plants to respond dynamically to environmental cues and optimize their survival strategies. (Abiri et al., 2017)

The study of gene duplication and evolutionary expansion of the ERF gene family provides valuable insights into the mechanisms underlying plant adaptation and stress tolerance. In sugarcane, understanding how duplicated ERF genes have evolved and diversified can help identify key regulatory genes associated with stress responses. Such knowledge is essential for developing improved crop varieties with enhanced tolerance to environmental stresses through molecular breeding and genetic engineering approaches. (Sakuma et al., 2002)

In the present study, a comparative genomics approach was employed to identify and characterize the ERF gene family in sugarcane and its wild relatives. Reference datasets from *Arabidopsis thaliana* and *Oryza sativa* were used to classify sugarcane ERF genes into subfamilies and subgroups. Phylogenetic analysis, conserved domain identification, motif analysis, protein characterization, and gene structure analysis were performed to understand the evolutionary relationships and functional diversity of these genes. Chromosomal mapping further provided insights into their genomic distribution. (Nakano et al., 2006)

The findings of this study are expected to provide a comprehensive understanding of the ERF gene family in sugarcane and their role in stress responses. By identifying and classifying ERF genes based on their evolutionary relationships with model plants, this study enables functional prediction of sugarcane genes. Furthermore, the identification of candidate genes associated with stress tolerance can contribute to the development of improved sugarcane varieties through molecular breeding and genetic engineering approaches. (Xu et al., 2006).

2. AIM AND OBJECTIVE:

Aim

To perform Comparative genomics and evolutionary analysis of the ERF gene family in sugarcane and its wild relatives

Objectives

- To identify and characterize the ERF gene family in sugarcane and its wild relatives using comparative genomics approaches.
- To investigate synteny relationships, gene duplication patterns, and evolutionary dynamics underlying the expansion and diversification of ERF genes in polyploid sugarcane.

3. REVIEW OF LITERATURE

3.1 Older Studies

3.1.1 The Sugarcane Genome

The biological story of sugarcane (*Saccharum* spp.) is one of extreme genomic ambition and complexity. While most major food crops like *Oryza sativa* or wheat have relatively straightforward genetic blueprints, sugarcane exists as a "mega-polyploid" mosaic. To understand why a transcription factor family like AP2/ERF is so massive in this crop, we must first look at the historical process of "nobilization" that defined modern cultivars (Garsmeur et al., 2018).

Breeders in the late 19th and early 20th centuries were faced with a dilemma: the high-sugar "noble" canes (*S. officinarum*) were physically weak and prone to disease, while the wild relatives (*S. spontaneum*) were virtually indestructible but produced little sugar. By crossing these two species, breeders essentially fused two different evolutionary strategies. In a modern hybrid like R570, this results in a genomic landscape where approximately 80% of the chromosomes come from the noble parent, while the wild parent contributes the essential "survival" genes—roughly 10-15% of the total genome (Li et al., 2020). For a researcher, this means that every single gene can have eight to twelve different versions (homeologs) across the sub-genomes. This redundancy is likely what allows sugarcane to withstand harsh environments, but it makes the task of identifying specific stress-response regulators like the AP2/ERF family a massive computational undertaking.

3.1.2 The AP2/ERF Superfamily

The discovery of the APETALA2/Ethylene Responsive Factor (AP2/ERF) superfamily in the 1990s marked a turning point in our understanding of how plants "think" and react. It began with the isolation of the *APETALA2* gene by Jofuku et al. (1994), which showed how plants coordinate the intricate development of flowers and seeds. Almost simultaneously, Ohme-Takagi and Shinshi (1995) identified the Ethylene Response Factors (ERFs) as the bridge between the hormone ethylene and the plant's actual physical defense responses.

What makes this family so unique is its structural "signature"—the AP2 domain. This sequence of 60 to 70 amino acids folds into three anti-parallel β -sheets and one α -helix, a shape that allows it to lock onto DNA with incredible precision (Allen et al., 1998). Over millions of years, this single domain has diversified into several specialized "tools" or subfamilies .

The AP2 Subfamily usually possesses two domains; these genes are the planners of the plant's physical structure. The ERF Subfamily act on ethylene signals to fight off pathogens and environmental shocks. The DREB Subfamily are specifically tuned to bind the DRE/CRT element (CCGAC), which is the molecular equivalent of a "red alert" for drought and freezing (Yamaguchi-Shinozaki and Shinozaki, 1994). The RAV and Soloists are smaller groups that integrate different hormone signals, ensuring that the plant's growth doesn't clash with its stress responses.

3.1.3 Group VII ERFs and the Waterlogging

Among the twelve subgroups of the ERF family, Group VII (ERF-VII) represents one of the most elegant examples of biological engineering. In sugarcane-growing regions, sudden monsoons often lead to waterlogging. When roots are submerged, they effectively begin to "drown" as oxygen levels drop (hypoxia). The plant cannot wait for a slow genomic response; it needs a sensor that reacts instantly.

This is where the ERF-VII genes come in. They are governed by the Arg/N-end rule pathway, a biochemical mechanism that acts like a countdown clock (Giuntoli and Perata, 2018). Under normal conditions, oxygen is present, and these proteins are systematically destroyed by the cell as soon as they are made. This seems wasteful, but it is actually a high-speed standby mode. The moment waterlogging occurs, and oxygen disappears, the "destruction signal" is cut off. The ERF-VII proteins suddenly stabilize, flood the nucleus, and switch on Hypoxia-Responsive Genes (HRGs) like *Alcohol Dehydrogenase (ADH)* to keep the plant's energy levels up through fermentation (Gasch et al., 2016; Ma et al., 2024). In *Oryza sativa*, this mechanism allows varieties like *SUB1A* to survive weeks underwater (van Veen et al., 2014). For sugarcane researchers, identifying these specific Group VII genes across the complex R570 and *S. officinarum* genomes is the "holy grail" for breeding crops that can survive the increasing unpredictability of global weather patterns.

3.2 Recent Advances in AP2/ERF Research

The landscape of sugarcane research has shifted dramatically with the transition from traditional genetics to "pangenomics." Recent studies have moved beyond simply identifying genes in a single species to comparing how the AP2/ERF superfamily has diversified across different *Saccharum* lineages. This is particularly vital because the regulatory requirements of a high-sugar "noble" cane (*S. officinarum*) differ vastly from those of a stress-hardy wild relative (*S. spontaneum*).

3.2.1 High-Resolution Identification in Polyploid Backgrounds

A breakthrough occurred with the release of the allele-defined genome assembly of *S. spontaneum*. As documented by Li et al. (2020), the identification of 218 AP2/ERF genes provided the first evidence of how whole-genome duplication (WGD) drives the expansion of stress-responsive families. Recent investigations into commercial hybrids like R570 have further expanded this, revealing that the superfamily can exceed 1,000 members when considering all homeologous copies across the sub-genomes. These recent studies emphasize that the "expansion" of the ERF subfamily is not merely a numbers game; it is a strategic evolutionary adaptation. By having multiple copies of a single transcription factor, sugarcane can "fine-tune" its response—using one version of a gene for mild drought and another for extreme desiccation (Garsmeur et al., 2018; Ma et al., 2024).

3.2.2 Structural Divergence of the AP2 Domain

Recent structural biology advances have revealed that the AP2 domain is far more dynamic than previously thought. While the "standard" model includes three β -sheets and one α -helix, recent deep-sequencing data has identified "non-canonical" AP2 domains in sugarcane. These variants often possess substitutions in the highly conserved 14th Valine and 19th Glutamic Acid residues. Research suggests that these subtle mutations allow certain sugarcane ERFs to bind to a broader range of *cis*-acting elements beyond the traditional GCC-box, effectively widening the plant's "regulatory net" during multi-stress events, such as the simultaneous occurrence of heat and salinity (Licausi et al., 2013; Li et al., 2020).

3.2.3 The Evolution of the N-terminal Motif in Group VII ERFs

The most specific recent advance concerning waterlogging is the identification of the MCGGAI/L motif in Group VII ERFs. While older studies identified this motif in *Arabidopsis*, recent comparative genomics across C4 grasses like sugarcane and sorghum has shown that this "oxygen-sensing tail" is remarkably conserved but often accompanied by additional downstream motifs. These auxiliary motifs, discovered through de novo discovery tools like MEME, are thought to interact with co-repressors or co-activators that are specific to the C4 photosynthetic pathway.

Current literature (2021–2024) suggests that in sugarcane, these ERF-VIIIs don't just regulate anaerobic metabolism; they also coordinate the "shut down" of sucrose transport during floods to conserve energy (Gasch et al., 2016). This functional crosstalk between the oxygen-sensing

machinery and the sugar-loading mechanisms is a cutting-edge area of research that was virtually unknown a decade ago.

3.2.4 Insights into "Intronless" Gene Architecture

One of the most striking recent observations in the structural analysis of the AP2/ERF family is the prevalence of intronless genes. In *S. spontaneum*, nearly 42% of the identified AP2/ERF genes lack introns (Li et al., 2020). Recent evolutionary theories propose that this is an "emergency response" architecture. Because these genes do not require the time-consuming process of pre-mRNA splicing, they can be transcribed and translated with near-instantaneous speed. This allows sugarcane to mount a "first-wave" defense within minutes of a stress trigger—such as the sudden onset of a tropical rainstorm that leads to root zone saturation.

3.2.5 Hormonal Crosstalk and the Soloist Enigma

Finally, recent advances have begun to solve the mystery of the "Soloist" subfamily. Long considered an evolutionary oddity, recent transcriptomic profiling in sugarcane has shown that Soloist genes, such as *SsSoloist4*, act as critical nodes in the crosstalk between Abscissic Acid (ABA) and Jasmonic Acid (JA) signaling. By responding to both drought and wounding signals, these genes ensure that the plant's defense resources are distributed efficiently. This "integrated signaling" model is the newest paradigm in transcription factor research, moving away from the old "one gene, one stress" philosophy toward a more holistic "regulatory network" view (Ma et al., 2024).

3.3 Bioinformatics Methodologies in AP2/ERF Characterization

The "bioinformatics revolution" has fundamentally shifted plant research from the era of single-gene studies to the systematic, family-wide characterization of entire regulatory networks. When dealing with a crop as genetically daunting as sugarcane—specifically the R570 (1020 genes), *S. officinarum* (1336 genes), and *S. spontaneum* (176 genes)—traditional web-based tools are no longer sufficient. The extreme polyploidy and high allele frequency of these genomes necessitate a high-performance computational pipeline that is both scalable and reproducible. By moving the workflow into a Linux-based Command Line Interface (CLI), we were able to process thousands of sequences simultaneously, ensuring that every gene was scrutinized with identical stringency regardless of the dataset's size.

3.3.1 The Linux CLI and HMMER

The reliance on the Linux Command Line Interface (CLI) served as the primary differentiator in this study's technical execution. Unlike Graphical User Interfaces (GUIs), which are often limited by memory overhead and lack of automation, the CLI allowed for the execution of high-throughput shell scripts. This was essential for the initial identification phase using HMMER 3.3.2. HMMER is a sophisticated search tool that utilizes Hidden Markov Models to build a probabilistic "profile" of a protein family. Rather than searching for a simple string of amino acids, HMMER looks for the conserved structural "mathematical consensus" of the AP2 domain (PF00847).

In the diverse genetic background of sugarcane, where domains can shift slightly between sub-genomes, this profile-based search was the only way to ensure that no divergent or non-canonical AP2/ERF members were overlooked. This mirrors the methodology used by Li et al. (2020) in their landmark study of *S. spontaneum*, where HMMER was used to successfully navigate the complex 3.36 Gb genome to identify 218 foundational AP2/ERF members. By adopting this same high-performance standard, this study was able to scale the identification to the even more complex hybrid backgrounds of R570 and *S. officinarum*.

3.3.2 Classification Logic and iTAK Adjudication

Once the raw pool of candidates was identified, the study utilized iTAK v1.3 to perform a rigorous "judicial" classification. iTAK is a specialized plant transcription factor toolkit that uses a consensus of multiple HMM profiles to sort sequences into their respective subfamilies, such as ERF, DREB, and RAV. This step was critical because the sugarcane genome often contains "noise"—sequences that may mimic the AP2 domain but lack the functional residues required for actual DNA binding. iTAK served to adjudicate these sequences, ensuring that only true regulatory factors remained in the final dataset.

This tiered approach to classification is now the industry standard for polyploid crops. For instance, Das et al. (2021) highlighted that using iTAK in conjunction with local databases is essential for eliminating false positives in large-scale plant proteomes. To further refine this, a Local BLASTp was executed within the Linux environment. This permitted the identification of high-identity paralogs across the R570 and *S. officinarum* genomes, followed by a manual redundancy filter to ensure that the final counts of 1,020 and 1336 genes represented unique genetic loci rather than assembly artifacts.

3.3.3 Evolutionary Reconstruction via Clustal Omega and MEGA 12

To resolve the complex evolutionary history of these hundreds of genes, the study employed Clustal Omega for large-scale sequence alignment. Clustal Omega is technically superior for this task because it utilizes "seeded guide trees" and HMM-profile-profile alignments, which maintain accuracy even as the number of sequences grows into the thousands (Sievers et al., 2011). Following this alignment, the phylogenetic architecture was reconstructed using MEGA 12 via the Maximum Likelihood (ML) method.

The ML approach is the gold standard for large families like AP2/ERF because it uses statistical inference to account for varying rates of evolution across different amino acid sites. This approach was similarly used by Yao et al. (2020) to organize the 317 AP2/ERF genes in hexaploid wheat, in where ML trees revealed the retention of homeologs across the A, B, and D subgenomes. By applying the Jones-Taylor-Thornton (JTT) model, the computer accounts for the varying rates of amino acid mutations across different sites. To ensure the "branches" of the tree were mathematically sound, we performed 500 bootstrap replicates. This statistical re-sampling ensures that the clustering of the 1,020 R570 genes into specific clades is backed by high confidence levels. The resulting Newick (.nwk) files were then projected onto the Interactive Tree of Life (iTOL), creating a spatial map that aligns the sugarcane isolates with the established regulatory groups of *Arabidopsis* and *Oryza sativa* (Letunic & Bork, 2021)

3.3.4 Structural Synthesis: MEME, NCBI-CDD, and TBtools

The most critical phase of the bioinformatics pipeline involved the synthesis of structural and evolutionary data using TBtools. Developed by Chen et al. (2020), TBtools acts as an integrative bridge, allowing for the simultaneous visualization of multiple datasets. To confirm the functional "machinery" of each protein, the study performed *de novo* motif discovery using the MEME Suite, identifying up to 15 unique signatures—including the EAR-repression motifs and oxygen-sensing tails. Simultaneously, every protein was passed through the NCBI Conserved Domain Database (CDD) to verify the structural integrity of the AP2 scaffold (Marchler-Bauer et al., 2017).

By importing both the MEME and CDD results into TBtools and aligning them alongside the phylogenetic tree, we created "Integrated Structural Maps." This technique has become increasingly popular in "Big Data" biology, as it allows researchers to correlate phylogenetic clusters with specific physical architectures. For example, similar integrated mapping in cultivated lettuce (*Lactuca sativa*) helped identify how tandem duplications drove the

expansion of specific stress-responsive clades (Park et al., 2023). In our study, this was the only way to prove that the 1336 genes of *S. officinarum* were not just random duplicates, but organized groups of proteins sharing identical functional machinery.

3.3.5 Physical Mapping and Synteny Analysis: GSDS and Chromosomal Localization

The final methodological step involved mapping the gene architecture and its physical location on the sugarcane chromosomes. Using the Gene Structure Display Server function within TBtools, performed Intron-Exon mapping. This required the preparation of precise chromosomal coordinates and gene position files via the Linux command line. This analysis is vital for understanding transcriptional speed; Hu et al. (2015) established that genes lacking introns—which we found in over 40 percent of the sugarcane ERF family—are often the first to respond to sudden environmental shocks because they do not require time-consuming mRNA splicing. To manage the visualization of over a thousand genes, a systematic renaming strategy was implemented. This logic was recently applied in the analysis of the *Brassica napus* genome to prevent visual clustering from making the data unreadable (Song et al., 2016). By mapping these renamed IDs against the chromosomal scaffolds, the study identified specific duplication hotspots, revealing how the noble *S. officinarum* expanded its genetic repertoire so much more aggressively than the wild *S. spontaneum*.

To decode the mechanism behind the 7.5-fold expansion, Synteny Analysis was performed using MCScanX. This revealed that the ERF family growth was primarily driven by Segmental and Whole Genome Duplication events. By identifying highly conserved syntenic blocks, the study traced the high-fidelity inheritance of stress-response clusters from the wild ancestor to the Hybrid R570. This evolutionary retention confirms that the modern hybrid maintains an expanded dosage-buffering capacity, providing a strategic genomic roadmap for targeting waterlogging-tolerance regions in complex sugarcane breeding. Despite the exponential growth of genomic data in the last decade, a critical disconnect persists between the theoretical knowledge of model systems and the practical application in complex polyploid crops like sugarcane. While the AP2/ERF superfamily has been mapped with high precision in *Arabidopsis* and *Oryza sativa*, the transition to sugarcane represents a frontier of complexity that remains largely unmapped. The following gaps in the current literature form the scientific foundation and necessity for this thesis.

3.4 Research Gap Identified

3.4.1 Uncharacterized ERF-VIIs in Sugarcane

The most glaring deficiency in current sugarcane research is the total lack of functional data regarding Group VII ERF transcription factors. In other major cereals, these genes have been hailed as the " switches" for surviving anaerobic conditions. Engineering successes in barley (Mendiondo et al., 2016), wheat (Wei et al., 2019), and rapeseed (Zubrycka et al., 2023) have proven that stabilizing these oxygen-sensing proteins can save up to 80% of crop biomass during prolonged flooding.

In contrast, sugarcane—a crop that frequently faces 20–50% yield losses in flood-prone tropical lowlands—has no specific characterization of its ERF-VII members. Given that monsoons can submerge sugarcane root zones for up to 21 days, the absence of a molecular "blueprint" for how sugarcane manages this hypoxia is a profound omission in the field of crop improvement. This thesis addresses this by specifically identifying and classifying the ERF-VII candidates within the complex hybrid background, providing the first essential step toward flood-tolerant breeding.

3.4.2 Genomic Myopia: Beyond *S. spontaneum*

While the work of Li et al. (2020) provided a landmark catalog of 218 AP2/ERF genes in the wild relative *S. spontaneum* (AP85-441), the scientific community has largely ignored the more complex commercial and ancestral genotypes. There is currently no comparable phylogenetic, chromosomal, or structural mapping for the "noble" cane *S. officinarum* or the elite commercial hybrid R570.

The biological reality is that a wild ancestor's genome is not a perfect mirror for a modern hybrid with over 115 chromosomes. By restricting research to a single wild accession, we lose the ability to see how these gene families expanded and specialized during the "nobilization" process. This thesis fills this "genomic blind spot" by performing a comparative analysis across three distinct genomic backgrounds, allowing us to see for the first time how the AP2/ERF family differs between a wild ancestor, a noble ancestor, and a modern high-yielding cultivar.

3.4.3 The Need for a Centralized, Multi-Dimensional Resource

Finally, there is a distinct lack of an integrated, "one-stop" resource for sugarcane AP2/ERF data. Currently, a researcher looking for information must piece together data from fragmented

databases that often lack consistency. There is no centralized resource that combines robust phylogenetic classification with the specific physical details of these genes—such as their intron-exon organization, physicochemical properties, and subcellular localization predictions.

As established by Chen et al. (2020) and Letunic & Bork (2021), the modern standard for genomics is "Data Synthesis." A phylogenetic tree is only useful if it is linked to structural motifs and chromosomal hotspots. This thesis adopts this "Synthesis Model," benchmarked against the gold-standard reference frameworks of *Arabidopsis thaliana* and *Oryza sativa*. By using a high-performance Linux-based pipeline involving HMMER3, IQ-TREE2, MEME, and TBtools, this study creates a comprehensive molecular atlas. This atlas is not just a list of genes; it is a functional roadmap that future breeders can use to identify which specific genetic loci are the best targets for CRISPR-mediated editing or marker-assisted selection.

4. MATERIALS AND METHOD

4.1. Computational Architecture and Hardware Environment

The execution of the high-throughput bioinformatics pipeline required a stable, dedicated hardware environment to manage the large-scale genomic datasets for *Saccharum*. The technical specifications of the workstation used for sequence retrieval, HMMER searches, and phylogenetic reconstruction are detailed in Table 3.1.

Table 4.1: Hardware and System Specifications for Genomic Auditing

Component	Specification
Processor	Intel(R) Core (TM) i5-6500 CPU @ 3.20 GHz
Installed Memory (RAM)	12.0 GB DDR4
Graphics Processing	NVIDIA GeForce (4 GB Dedicated
Storage Architecture	1.13 TB Hybrid Configuration (SSD/HDD)
Operating System	Windows 10 Pro (Version 22H2)
System Architecture	64-bit OS, x64-based Processor

4.2. Software Configuration

Ubuntu 20.04 LTS (via Windows Subsystem for Linux - WSL): Used as the high-performance computing environment for executing Command Line Interface (CLI) operations and managing large-scale genomic datasets.

MEGA 12 (Molecular Evolutionary Genetics Analysis): The primary engine for multiple sequence alignment (MSA) and the construction of Maximum Likelihood (ML) phylogenetic trees using the JTT model and 500 bootstrap replicates.

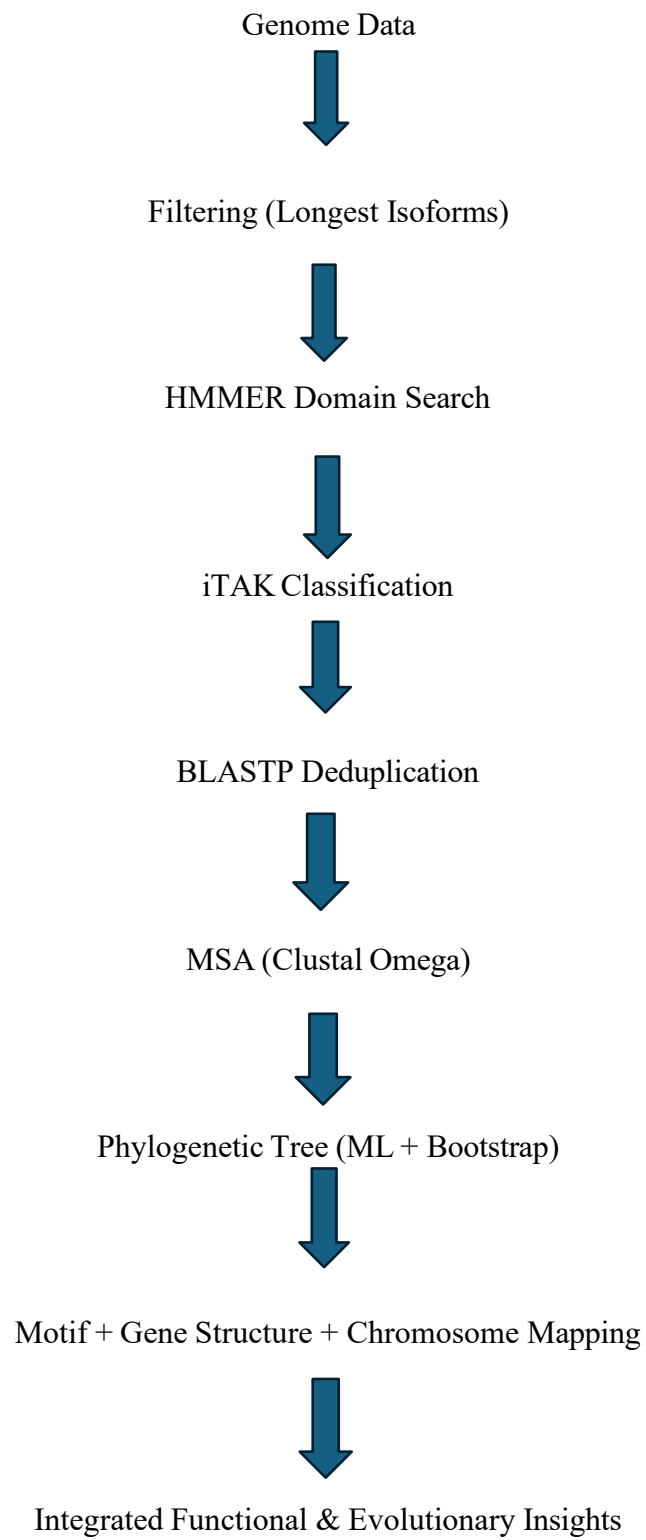
TBtools v1.09 (Toolkit for Biologists): An integrative software suite used for chromosomal localization, Gene Structure Display (GSDS), motif visualization, and synteny mapping.

BLAST+ Standalone Suite (NCBI): Utilized for local, high-speed sequence similarity searches to identify homeologs and paralogs within the sugarcane proteomes.

HMMER v3.3.2: A profile-based search tool installed within the Ubuntu environment to identify the conserved AP2 domain (PF00847) using Hidden Markov Models.

Microsoft Excel: Used for the systematic management of gene

4.3 Methodological Workflow



4.4 Bioinformatic Methodology and High-Throughput Data Processing

4.4.1 Genomic Resource Acquisition and Dataset Preparation

A multi-genome data acquisition strategy was implemented to construct a high-resolution proteomic library, serving as the primary substrate for the genome-wide identification of the AP2/ERF superfamily. Given the extreme polyploidy and nested chromosome architecture of the *Saccharum* genus, we prioritized datasets with verified assembly metrics. The complete genome sequences, structural gene annotations (GFF3), and proteome FASTA files for three representative lineages—*S. officinarum*, the wild ancestor *S. spontaneum*, and the commercial hybrid R570—were retrieved from the Sugarcane Genome Hub (SGH). This centralized repository was selected as the primary source to ensure internal consistency in nomenclature and to mitigate the "bioinformatic noise" often encountered when integrating data from fragmented public archives.

To move beyond a localized catalog and provide an evolutionary "verdict," we integrated two gold-standard model systems as comparative benchmarks. A total of 163 AP2/ERF genes from *Oryza sativa* and 139 genes from *Arabidopsis thaliana* were retrieved from the Plant Transcription Factor Database (PlantTFDB). These reference sets were supplemented with full proteome downloads from Ensembl Plants and Phytozome v13. By utilizing both a monocot (*Oryza sativa*) and a dicot (*Arabidopsis*) anchor, we established a robust framework for tracing lineage-specific expansions and identifying conserved stress-response signatures across the sugarcane sub-genomes.

4.4.2 Domain-Centric Identification via Probabilistic Modeling (HMMER)

The identification of the ERF superfamily was conducted at the protein level to ensure functional accuracy. Unlike nucleotide-based searches, protein-level audits are resilient to synonymous mutations and codon usage bias, allowing for a more sensitive detection of the conserved 60–70 amino acid AP2 scaffold.

We executed a profile-based Hidden Markov Model search using the HMMER v3.3.2 software suite. The conserved AP2/ERF DNA-binding domain (PF00847) was utilized as the "mathematical query" to scan the three sugarcane proteomes. This probabilistic approach is superior to conventional BLAST-based similarity searches, as it can identify evolutionarily divergent homologs that maintain the structural fold of the domain despite low primary sequence identity. To balance sensitivity with biological relevance, we applied a stringent E-

value threshold of < 0.01 . This statistical filter served as the initial "Quality Control" gate, retaining statistically significant candidates while discarding unrelated peptide fragments.

4.4.2.1 Multi-Layered Structural Validation (CDD and Pfam Adjudication)

The raw output from the HMMER search, while comprehensive, often contains non-canonical proteins or fragmented domain architectures. To resolve this, we implemented a dual-validation "Adjudication" strategy using the **NCBI Conserved Domain Database (CDD)** and **Pfam**. Each candidate was individually audited to verify the structural integrity of the AP2 domain.

This two-step verification was essential for distinguishing between true transcription factors and proteins with incomplete or "broken" domains. Only sequences exhibiting a complete, canonical domain architecture were retained for downstream analysis. We specifically scrutinized the sequences to categorize them into three structural archetypes:

- AP2 Family: Characterized by tandem AP2 domain repeats.
- ERF/DREB Family: Defined by a solitary, high-affinity AP2 domain.
- RAV Family: Distinguished by a chimeric AP2 + B3 domain architecture.

Sequences exhibiting ambiguous or truncated motifs were purged from the dataset to prevent phylogenetic artifacts.

4.4.2.2 Redundancy Management and Paralogous Filtering (Local BLAST+)

A significant challenge in sugarcane genomics is the presence of high-identity homeologs and duplicated gene copies arising from recent whole-genome duplication (WGD) events. To address this, we executed a Local BLASTP analysis within a Ubuntu Linux environment. This "Self-to-Self" comparative audit was performed within each species' dataset to identify paralogous sequences.

To ensure that the final gene census represented unique genetic loci rather than assembly alleles, we applied a "Strict Homology Matrix" for filtering. Alignments were only considered significant if they met the following statistical stress-tests:

- Sequence Identity: $> 50\text{--}60\%$ (to capture broad paralogy).
- Query Coverage: $> 70\%$ (to ensure structural homology).
- E-value: $< 1 \times 10^{-5}$ (for statistical confidence).

Furthermore, we implemented a Redundancy Elimination Protocol. Duplicate sequences sharing greater than 95% identity across overlapping regions—often representing alternative gene models—were reduced to a single representative entry. This rigorous pruning was technically critical to ensure that our phylogenetic models were not "drowned out" by identical sequences, allowing for a clear visual representation of the evolutionary clades.

4.4.2.3. Final Census and Comparative Dataset Integration

Through this multi-layered filtering pipeline, we established a high-confidence, non-redundant repertoire for the three sugarcane lineages. The finalized audit revealed a significant disparity in gene density, reflecting the complex ploidy levels of the targets:

- Modern Hybrid (R570): 1,020 high-confidence ERF genes.
- Noble Sugarcane (*S. officinarum*): 1336 high-confidence ERF genes.
- Wild Ancestor (*S. spontaneum*): 176 high-confidence ERF genes.

Finally, to facilitate comparative mapping, these datasets were merged using the TBtools Sequence Toolkit. We generated six input files, pairing each sugarcane species separately with the *Arabidopsis* and *Oryza* reference sets. This systematic integration ensured that naming conventions and sequence formats were perfectly synchronized, providing the necessary structural framework for the subsequent Phylogenetic Reconstruction and Motif Identification phases.

4.4.3 Multiple Sequence Alignment and Evolutionary Reconstruction

To resolve the complex divergence and expansion of the AP2/ERF superfamily across the three *Saccharum* lineages, a structured comparative phylogenetics framework was implemented. The primary challenge posed by the dataset's scale—comprising 1,020 (R570), 1336 (*S. officinarum*), and 176 (*S. spontaneum*) loci—necessitated a "Global-to-Local" benchmark strategy. Six independent datasets were synthesized, integrating each sugarcane proteome with established reference sequences from *Arabidopsis thaliana* (139 genes) and *Oryza sativa* (163 genes). This integration provided a stable taxonomic scaffold, allowing for the classification of sugarcane homeologs into well-defined botanical subfamilies based on orthologous proximity (Nakano et al., 2006).

4.4.3.1. High-Throughput Multiple Sequence Alignment (MSA)

The fundamental prerequisite for accurate evolutionary modeling is the maximization of positional homology. Given the massive number of sequences, Clustal Omega was executed via the Linux Command Line Interface (CLI) to generate the Multiple Sequence Alignment (MSA). Unlike traditional progressive aligners, Clustal Omega utilizes a Hidden Markov Model (HMM)-based strategy, which is uniquely optimized for handling datasets exceeding 1,000 sequences (Sievers et al., 2011). This approach ensured that the conserved 60–70 amino acid AP2 scaffold remained aligned across all six datasets, while accurately capturing the variation in the flanking C-terminal and N-terminal regions. The resulting alignments provided a "Strict Homology Matrix," essential for reducing the topological noise in subsequent tree construction.

4.4.3.2. Maximum Likelihood (ML) Modeling and Statistical Stress-Testing

The evolutionary "verdict" for the superfamily was delivered using the Maximum Likelihood (ML) method within the MEGA 12 (Molecular Evolutionary Genetics Analysis) suite. The ML approach was selected for its statistical superiority in evaluating multiple tree topologies to identify the most probable evolutionary trajectory of the sugarcane sequences (Tamura et al., 2021).

To account for the specific empirical rates of amino acid substitution within complex plant proteomes, the Jones-Taylor-Thornton (JTT) model was applied. This substitution matrix is critical for analyzing conserved transcription factor families, as it mathematically corrects for the differential rates of transitions and transversions at the peptide level. To ensure the reliability of the resulting clades, a Bootstrap Analysis with 500 replicates was performed. This statistical re-sampling process provides a confidence interval for each internal node; branches with high bootstrap support were interpreted as stable evolutionary lineages, while lower values indicated areas of rapid recent expansion or genetic drift within the sugarcane sub-genomes (Felsenstein, 1985; Kumar et al., 2018).

4.4.3.3. Topological Partitioning and Clade Classification (iTOL)

The finalized phylogenetic trees generated from the Maximum Likelihood analysis were exported in Newick format and subsequently uploaded to the Interactive Tree of Life (iTOL v6) platform for detailed visualization, annotation, and comparative interpretation. This platform enabled effective handling of large and complex tree structures, allowing the

integration of multiple datasets and clear differentiation of species-specific gene clusters. Within iTOL, the ERF genes from the three sugarcane species were systematically annotated alongside reference sequences from *Arabidopsis thaliana* and *Oryza sativa*, and the tree was visually enhanced through strategic color-coding and labeling to improve interpretability.

Rather than relying solely on broad subfamily classification (such as ERF and DREB), the study refined a subgroup-based classification approach. Specifically, the classification framework proposed by Nakano et al. (2006), which is based on a genome-wide analysis of ERF genes in *Arabidopsis* and *Oryza sativa*, was used as the primary reference for categorization. Based on this system, the sugarcane ERF genes were assigned to distinct subgroups ranging from Groups I to X and, where applicable, extended classifications up to Groups I to XV, depending on their phylogenetic clustering patterns. The classification was performed by carefully examining the positioning of sugarcane sequences relative to well-characterized reference genes within the tree. Accordingly, the phylogenetic tree was color-coded based on these subgroup assignments, allowing clear visualization of subgroup distribution and evolutionary relationships across species.

This subgroup-oriented classification provided a more detailed and functionally meaningful framework, as individual groups are often associated with specific regulatory roles. For instance, certain clusters corresponding to dehydration-responsive elements (DREB-like groups) and stress-related regulatory pathways could be identified based on their alignment with known reference sequences. By focusing on subgroup classification rather than generalized subfamily division, a more precise mapping of gene function and evolutionary conservation was achieved.

In addition to the primary ERF clusters, sequences exhibiting distinct structural features were also examined to identify members of other AP2/ERF superfamily branches. Genes containing tandem AP2 domains were noted and interpreted separately in accordance with established classification systems, allowing distinction of AP2-like lineages where present. Similarly, highly divergent sequences and those associated with additional conserved domains, such as the B3-domain-containing RAV proteins and Soloist-like members, were identified as separate clusters or outgroups within the phylogenetic framework. These divergent groups were not included in the main ERF subgroup classification but were retained to provide evolutionary context and support the overall topology of the tree.

The visualization and subgroup-based annotation phase played a critical role in interpreting lineage-specific patterns within the sugarcane ERF gene family. By examining the distribution and clustering density of sugarcane genes within specific *Arabidopsis* and *Oryza sativa* subgroups, it was possible to identify patterns of expansion, conservation, and divergence among the three sugarcane species, particularly in *S. officinarum* and R570. This approach enabled the construction of a comprehensive evolutionary framework in which each identified ERF gene was not treated merely as an isolated sequence, but was instead associated with a predicted functional role based on its phylogenetic proximity to experimentally characterized genes from model plants. Thus, the subgroup-based classification strategy provided both structural clarity and functional insight into the organization of the sugarcane AP2/ERF superfamily.

4.4.5 Structural Auditing: Motif, Gene Structure, and Topology

To quantify the functional specialization and structural divergence of the identified AP2/ERF superfamily, a multi-dimensional architectural analysis was conducted. This phase sought to bridge the gap between primary sequence identification and the underlying genomic "logic" governing these transcription factors.

4.4.5.1. *De Novo* Motif Discovery and Peptide Signatures

Conserved protein signatures were identified using the MEME (Multiple Em for Motif Elicitation) Suite (v5.4.1). This analysis was specifically configured to detect a maximum of 10 unique motifs per dataset, providing a high-resolution map of both the conserved AP2 DNA-binding scaffold and the more variable regulatory "tails" (Bailey et al., 2015). By identifying recurring amino acid patterns, such as the EAR-repression motifs (L-X-L-X-L) or specific activation domains, we were able to predict the regulatory behavior of various ERF subgroups. This motif-based "fingerprinting" is essential for validating phylogenetic clustering, as members of the same clade (e.g., Groups I–X) typically exhibit near-identical motif architectures, reflecting shared evolutionary origins and functional roles in stress adaptation.

4.4.5.2 Intron–Exon Organization

The exon–intron organization of AP2/ERF genes was analyzed using TBtools (Chen et al., 2020). Gene structure visualization was performed by aligning coding sequences (CDS) with their corresponding genomic sequences. This analysis was conducted separately for each sugarcane species (*S. officinarum*, R570 hybrid, and *Saccharum spontaneum*) using genome-specific datasets, rather than merged datasets, to ensure accurate representation of

gene structures within each genome. Two input datasets were prepared for this analysis: a gene position file, containing gene IDs along with chromosome number, start position, and end position, and a chromosome length file, containing the total length of each chromosome. These datasets were used to generate graphical representations of exon–intron structures for each gene.

4.4.5.3 Chromosomal Localization

Chromosomal localization of AP2/ERF genes was performed using the Genome Location Mapper tool in TBtools. The analysis was carried out individually for each sugarcane genome, using species-specific gene position and chromosome length data. Merged datasets were not used for this analysis in order to preserve genome-level positional accuracy. Gene position data, including chromosome number and genomic coordinates, along with chromosome length information, were used as input files for visualization. For clarity in graphical representation, simplified gene identifiers (e.g., ScERF1, ScERF2, SsERF1) were used while maintaining correspondence with original gene annotations. The genes were mapped onto their respective chromosomal scaffolds to visualize their distribution across the genome.

4.4.6. Biophysical Profiling, Synteny, and Functional Validation

The final phase of the molecular audit focused on quantifying the physicochemical properties and the evolutionary forces driving gene family expansion.

4.4.6.1. Physicochemical and Subcellular Auditing

A comprehensive physicochemical profile for every identified ERF protein was generated using the ProtParam tool (ExPASy server). Calculated critical parameters, including Molecular Weight (MW) and the Theoretical Isoelectric Point (pI), to evaluate the biochemical stability and DNA-binding affinity of the proteins (Gasteiger et al., 2005). Simultaneously, subcellular localization was predicted via the CELLO2GO server. Given that AP2/ERF factors function as transcriptional switches, this step served as a functional validation; high-confidence nuclear localization signatures across the dataset confirmed that the identified sequences possess the necessary "molecular address" to operate within the nucleus (Yu et al., 2014).

4.4.6.2. Synteny Analysis and Duplication Logic

To identify the mechanisms driving superfamily expansion within the *Saccharum* complex, synteny and collinearity analyses were performed using MCScanX (Multiple Collinearity Scan

toolkit) integrated within the TBtools platform. This approach was employed to differentiate between various modes of gene duplication, specifically distinguishing large-scale Whole-Genome Duplications (WGD)/Segmental duplications from localized Tandem, Proximal, and Dispersed duplications (Wang et al., 2012).

Intraspecific synteny blocks were identified to trace homeologous gene pairs within *S. officinarum*, *S. spontaneum*, and the R570 hybrid. Additionally, interspecific collinearity maps were constructed between these *Saccharum* lineages and the diploid reference genome, *Sorghum bicolor*, to establish evolutionary conservation and ancestral duplication events. The resulting data were used to categorize the duplication history of each superfamily member, providing a statistically validated framework for interpreting the genomic architecture of the sugarcane AP2/ERF superfamily.

5. RESULT

5.1 Identification of AP2/ERF Candidate Genes in Sugarcane Genomes

The genome-wide identification of AP2/ERF transcription factors was carried out across three representative sugarcane lineages, namely *S. officinarum*, *Saccharum spontaneum*, and the commercial hybrid R570, along with reference datasets from *Arabidopsis thaliana* and *Oryza sativa*. Initially, a comprehensive proteome dataset was constructed using high-quality genome assemblies retrieved from the Sugarcane Genome Hub and other public repositories, thereby ensuring consistent gene annotation and sequence integrity (Zhang et al., 2018; Garsmeur et al., 2018).

To identify candidate AP2/ERF genes, a domain-centric approach was employed using the Hidden Markov Model (HMM) profile of the conserved AP2 DNA-binding domain (PF00847). The HMMER v3.3.2 tool was used to scan the entire proteome of each sugarcane genome, allowing the detection of both highly conserved and evolutionarily divergent members of the superfamily (Finn et al., 2011). This approach ensured high sensitivity in identifying true transcription factors, even in the complex polyploid background of sugarcane genomes (Riechmann and Meyerowitz, 1998).

The initial HMM search generated a large pool of candidate sequences; however, not all retrieved sequences represented functional AP2/ERF proteins. Therefore, a stringent filtering strategy was implemented. All candidate sequences were subjected to domain validation using the Conserved Domain Database (CDD) and Pfam to confirm the presence of a complete and intact AP2 domain (Marchler-Bauer et al., 2017; El-Gebali et al., 2019). Sequences lacking the conserved domain, containing truncated motifs, or representing partial gene models were excluded from further analysis to avoid inaccuracies in downstream interpretation.

To further refine the dataset, redundancy analysis was performed using local BLASTP (Altschul et al., 1997). Given the high level of genome duplication and presence of homeologous gene copies in sugarcane, sequences exhibiting high similarity (>95% identity) were considered redundant and collapsed into single representative entries. Additional filtering criteria, including sequence identity (>50–60%), query coverage (>70%), and E-value thresholds (<1e-5), were applied to ensure that only biologically relevant and structurally complete sequences were retained (Camacho et al., 2009).

Following this multi-layered validation and filtering pipeline, a final high-confidence set of AP2/ERF genes was established for each genome. The analysis revealed a substantial variation in gene family size across the three sugarcane lineages, reflecting their complex polyploid nature and evolutionary history (Zhang et al., 2018). Specifically, a total of approximately 1,020 genes in R570, 1,336 genes in *S. officinarum*, and 176 genes in *S. spontaneum* were identified as bona fide members of the AP2/ERF superfamily.

Additionally, reference AP2/ERF datasets comprising 163 genes from *Oryza sativa* and 139 genes from *Arabidopsis thaliana* were integrated into the study to provide a comparative evolutionary framework and support functional classification of the identified sugarcane genes (Nakano et al., 2006)

Table 5.1: Summary of AP2/ERF genes identified

Species	Number of Genes
<i>S. officinarum</i> (LA Purple)	1330
R570	1020
<i>S. spontaneum</i>	176

The distribution of AP2/ERF genes across the three sugarcane genomes shows a clear disparity in gene family size. *S. officinarum* (LA Purple) exhibits the highest number of genes, followed by R570, while *S. spontaneum* contains a comparatively smaller set. This pattern reflects the expansion of the AP2/ERF gene family in cultivated and hybrid lineages relative to the wild progenitor. The graphical representation (Figure 4.1) further emphasizes this trend, highlighting the differential gene abundance among the genomes. This expansion may be attributed to genome duplication and hybridisation events characteristic of cultivated sugarcane.

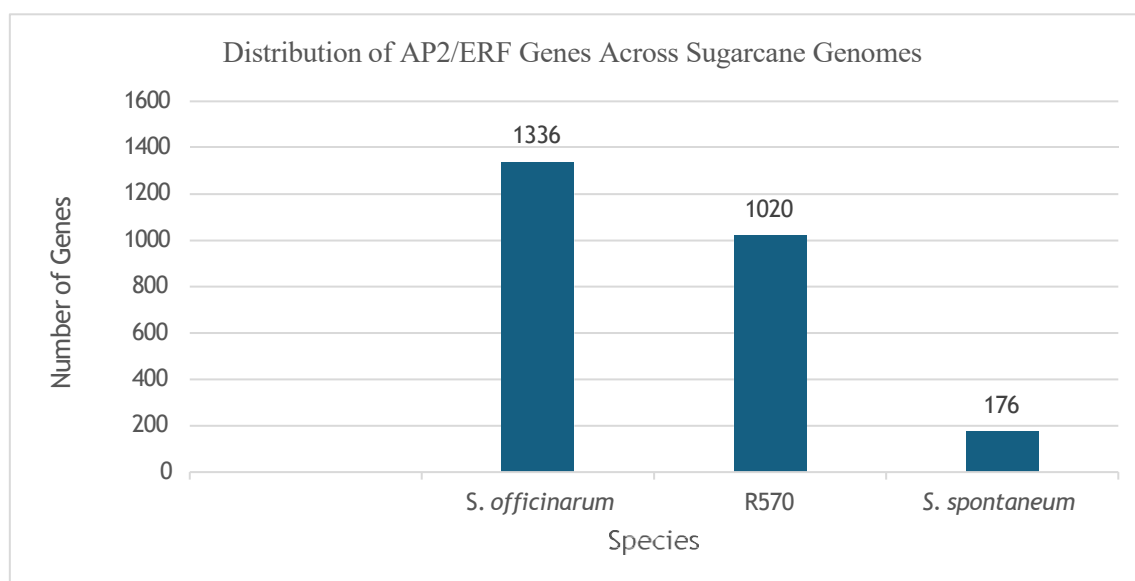


Figure 5.1: Distribution of AP2/ERF genes across three sugarcane genomes.

5.2 Domain Analysis of AP2/ERF Proteins

To further validate the identified AP2/ERF genes, domain analysis was performed using the NCBI Conserved Domain Database (CDD). Due to the large number of sequences, a comparative strategy was adopted by integrating sugarcane protein datasets with well-characterized reference AP2/ERF sequences from *Arabidopsis thaliana* (139 genes) and *Oryza sativa* (163 genes).

For this purpose, protein sequences from each sugarcane genome—*S. officinarum* (LA Purple), R570, and *S. spontaneum*—were separately merged with the reference datasets, generating a total of six input files.

These datasets were subjected to Batch CD-Search analysis using the NCBI Conserved Domain Database (CDD) to identify conserved domains and confirm the presence of the characteristic AP2 DNA-binding domain. This step served as a validation filter to ensure that only sequences with intact and functionally relevant domains were retained for downstream analysis.

5.2.1 *S. officinarum*

The domain analysis of *S. officinarum* confirmed a total of 1,336 AP2/ERF genes. In the merged datasets (*S. officinarum* + *Arabidopsis thaliana* and *S. officinarum* + *Oryza sativa*), the majority of sequences showed significant hits to the conserved AP2 DNA-binding domain (Pfam: PF00847; CDD: cd00018).

The identified domains exhibited strong statistical support, with E-values predominantly ranging from $1e^{-25}$ to $1e^{-34}$, indicating high confidence in domain assignment. Most proteins contained a single AP2 domain, characteristic of the ERF subfamily, while a smaller proportion displayed two AP2 domains (AP2 subfamily) or additional domains such as B3, corresponding to RAV members.

The domain organization remained consistent across both datasets, demonstrating that the inclusion of reference species did not affect the classification. These findings indicate a substantial expansion of structurally conserved AP2/ERF proteins in *S. officinarum*.



Figure 5.2 Conserved domain architecture of AP2/ERF proteins in *S. officinarum* in comparison with *Arabidopsis thaliana*, visualized using TBtools



Figure 5.3 Conserved domain architecture of AP2/ERF proteins in *S. officinarum* in comparison with *Oryza sativa*, visualized using TBtools

5.2.2 R570

In the hybrid cultivar R570, domain analysis validated 1,020 AP2/ERF genes. The merged datasets (R570 + *Arabidopsis thaliana* and R570 + *Oryza sativa*) consistently showed the presence of the conserved AP2 DNA-binding domain across the majority of sequences.

The proteins exhibited high-confidence domain hits, including matches to Pfam (PF00847) and CDD profiles (cd00018). Similar to *S. officinarum*, most sequences contained a single AP2 domain, indicating the predominance of the ERF subfamily, while comparatively fewer proteins exhibited dual AP2 domains or additional domains such as B3.

Despite the hybrid nature of the R570 genome, the domain architecture remained highly conserved, suggesting that gene expansion has retained functional integrity of AP2/ERF proteins.

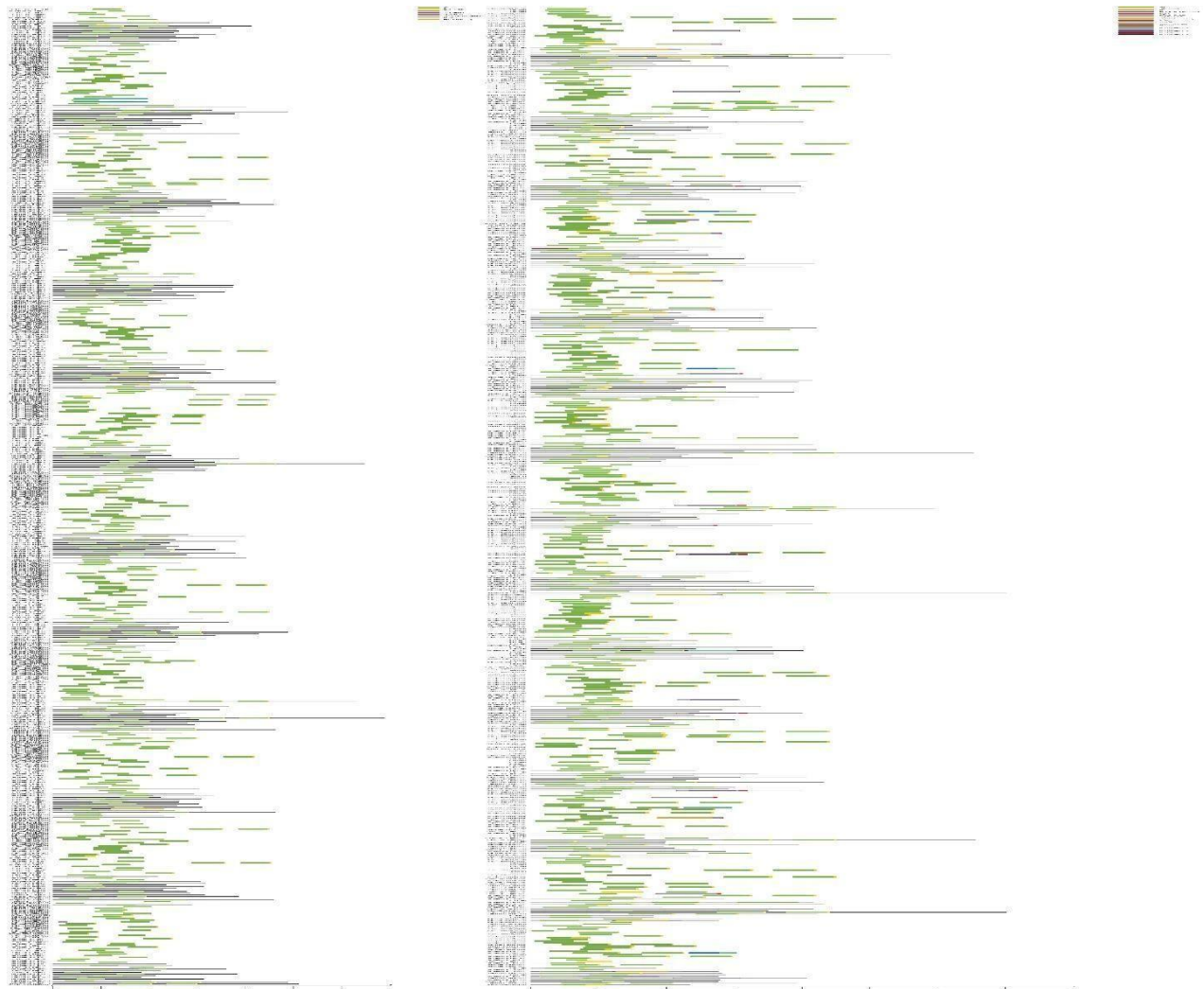


Figure 5.4 Conserved domain architecture of AP2/ERF proteins in R570 in comparison with *Arabidopsis thaliana*, visualized using TBtools



Figure 5.5 Conserved domain architecture of AP2/ERF proteins in R570 in comparison with *Oryza sativa*, visualized using TBtools

5.2.3 *Saccharum spontaneum*

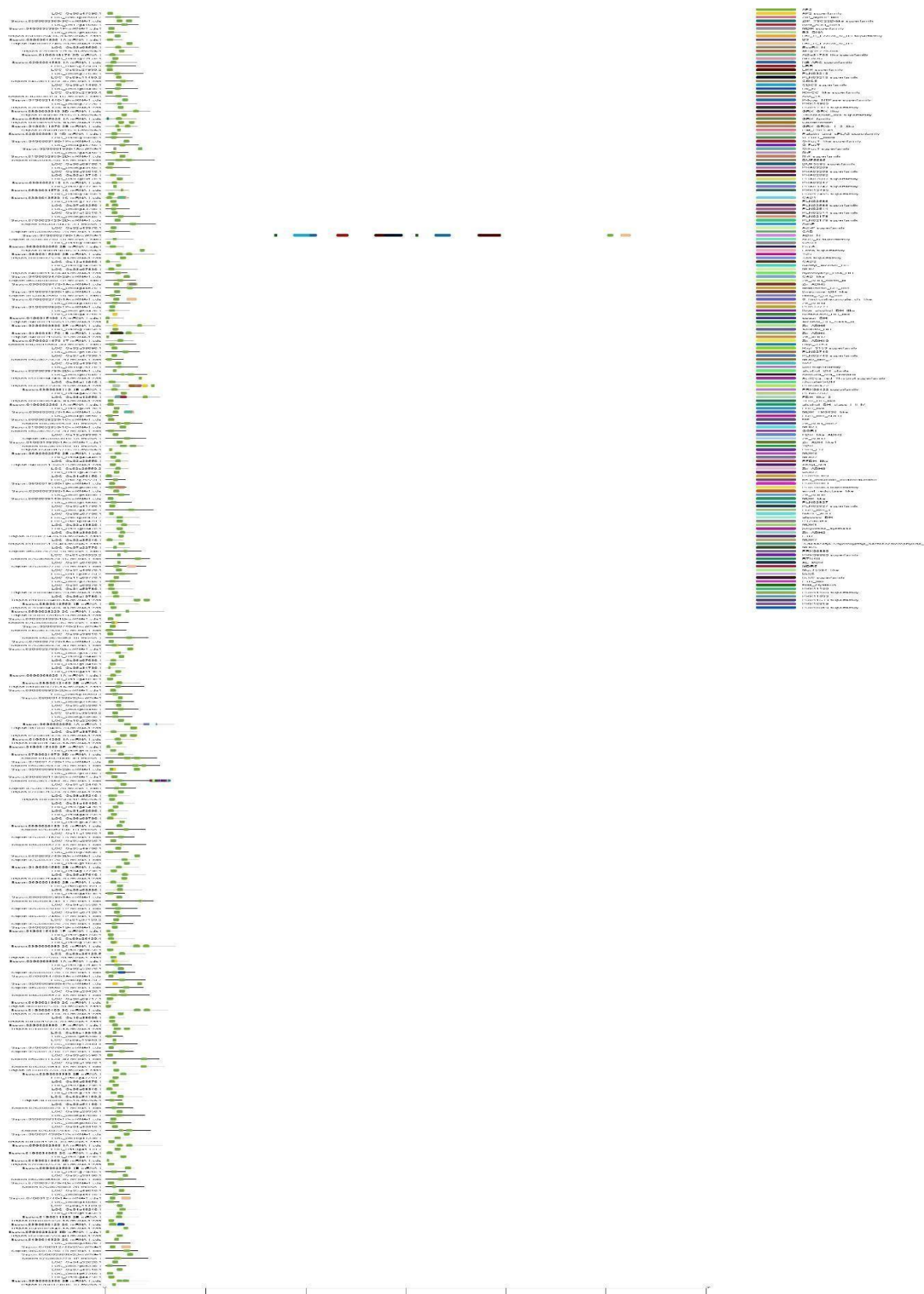
The domain analysis of *S. spontaneum* confirmed 176 AP2/ERF genes, representing a comparatively smaller gene set. In both merged datasets (*S. spontaneum* + *Arabidopsis thaliana* and *S. spontaneum* + *Oryza sativa*), the sequences showed clear and specific hits to the conserved AP2 DNA-binding domain.

The domain alignments demonstrated strong conservation, with most proteins containing a single AP2 domain, while fewer sequences belonged to AP2 and RAV subfamilies. Compared to the cultivated and hybrid genomes, *S. spontaneum* exhibited a more compact yet highly conserved domain architecture.

The conserved DNA-binding regions showed high similarity to those of both monocot and dicot reference species, reflecting its role as an ancestral lineage.



Figure 5.6 Conserved domain architecture of AP2/ERF proteins in *S. spontaneum* in comparison with *Arabidopsis thaliana*, visualized using TBtools



5.3 Motif Analysis

The conserved motif analysis of AP2/ERF proteins was carried out using the MEME Suite, with the maximum number of motifs set to 10. The identified motifs were further visualized using TBtools, which enabled the comparative analysis of motif distribution and organization across different datasets. Motif analysis provides insights into conserved functional regions beyond domain identification and helps in understanding the structural diversity and evolutionary relationships of transcription factors.

5.3.1 *S. officinarum*

The motif analysis of *S. officinarum* proteins was performed by integrating its sequences with those of *Arabidopsis thaliana* and *Oryza sativa*. The TBtools visualization of the MEME results revealed a highly organized and conserved motif distribution across the merged datasets.

In the merged dataset of the *S. officinarum* and *Arabidopsis thaliana* dataset, the majority of sequences exhibited a conserved set of core motifs, particularly Motifs 1, 2, and 3, which correspond to the AP2 DNA-binding domain. These motifs were consistently positioned across both species, indicating strong evolutionary conservation despite the monocot–dicot divergence. Additional motifs (Motifs 4–10) displayed subgroup-specific distribution patterns, contributing to functional diversification of the AP2/ERF proteins.

Similarly, in the *S. officinarum* and *Oryza sativa* dataset, an even higher level of motif conservation was observed. The motif arrangements showed strong collinearity between sugarcane and *Oryza sativa* sequences, reflecting their close evolutionary relationship as monocots. The core motifs (1–3) remained highly conserved in both position and structure, while auxiliary motifs (4–10) exhibited consistent patterns within specific clades.

Despite the large number of AP2/ERF genes identified in *S. officinarum*, the overall motif architecture remained highly stable across both datasets. This indicates that gene expansion in sugarcane has preserved the structural and functional integrity of transcription factors, resulting in a conserved yet expanded gene family.



Figure 5.8: Conserved motif distribution of AP2/ERF proteins in the merged dataset of *S. officinarum* and *Arabidopsis thaliana*, visualized using TBtools.



Figure 5.9: Conserved motif distribution of AP2/ERF proteins in the merged dataset of *S. officinarum* and *Oryza sativa*, visualized using TBtools

5.3.2 R570 Hybrid

The conserved motif analysis of the R570 hybrid AP2/ERF proteins was performed using the MEME Suite, with the maximum number of motifs set to 10. The identified motifs were visualized using TBtools to examine their distribution and organization across datasets. The analysis revealed a highly conserved and structured motif arrangement, indicating the preservation of functional protein architecture in the hybrid genome.

In the merged dataset of the R570 hybrid and *Arabidopsis thaliana*, the majority of sequences exhibited the conserved core motifs (Motifs 1, 2, and 3), corresponding to the AP2 DNA-binding domain. These motifs were consistently positioned across both species, indicating strong conservation of the DNA-binding region despite evolutionary divergence between monocots and dicots. Repetitive motif patterns observed among multiple R570 sequences suggest gene duplication and polyploid expansion. The auxiliary motifs (Motifs 4–10) displayed subgroup-specific distribution, contributing to the functional diversification of AP2/ERF proteins.

Similarly, in the merged dataset of the R570 hybrid and *Oryza sativa*, a higher degree of motif conservation was observed. The motif arrangements showed strong collinearity between R570 and *Oryza sativa* sequences, reflecting their close evolutionary relationship within the Poaceae family. Multiple R570 genes exhibited identical motif patterns corresponding to single *Oryza sativa* orthologs, indicating the presence of duplicated gene copies with conserved structural features. The core motifs (1–3) remained highly stable, while the auxiliary motifs (4–10) were consistently associated with specific clades.

Overall, the comparative analysis demonstrates that despite the hybrid and polyploid nature of the R570 genome, the motif architecture remains highly conserved. This suggests that gene expansion has retained the structural integrity and functional potential of AP2/ERF proteins, contributing to the regulatory complexity and adaptability of the hybrid sugarcane.



Figure 5.10: Conserved motif distribution of AP2/ERF proteins in the merged dataset of the R570 hybrid and *Arabidopsis thaliana*, as identified by MEME and visualized using TBtools.



Figure 5.11: Conserved motif distribution of AP2/ERF proteins in the merged dataset of the R570 hybrid and *Oryza sativa*, as identified by MEME and visualized using TBtools.

5.3.3 *S. spontaneum*

The conserved motif analysis of *S. spontaneum* AP2/ERF proteins was performed using the MEME Suite, with the maximum number of motifs set to 10. The identified motifs were visualized using TBtools to examine their distribution and organization across datasets. The analysis provided insights into the ancestral motif architecture of the AP2/ERF gene family.

In the merged dataset of *S. spontaneum* and *Arabidopsis thaliana*, the majority of sequences exhibited the conserved core motifs (Motifs 1, 2, and 3), corresponding to the AP2 DNA-binding domain. These motifs were consistently positioned across both species, indicating strong evolutionary conservation of the DNA-binding region. Compared to cultivated and hybrid sugarcane, the motif distribution in *S. spontaneum* appeared more streamlined, with distinct subgroup-specific arrangements of auxiliary motifs (Motifs 4–10). This clear organization suggests that the ancestral gene family possessed well-defined functional specialization with minimal redundancy.

Furthermore, specific motif patterns associated with functional groups were observed. The presence of conserved motif signatures related to transcriptional repression and activation indicates that key regulatory mechanisms were already established in the wild ancestor. These findings demonstrate that *S. spontaneum* provides the structural foundation for the expanded AP2/ERF gene family observed in modern sugarcane species.

Similarly, in the merged dataset of *S. spontaneum* and *Oryza sativa*, a high degree of motif conservation and collinearity was observed. The motif arrangements showed strong similarity between *S. spontaneum* and *Oryza sativa* sequences, reflecting their close evolutionary relationship within the Poaceae family. The core motifs (1–3) remained highly conserved, while the auxiliary motifs (4–10) exhibited consistent clade-specific patterns.

Unlike the polyploid genomes of *S. officinarum* and the R570 hybrid, the motif distribution in *S. spontaneum* showed a more compact and orthologous organization, indicating fewer duplicated gene copies. This suggests that the ancestral genome relied on structural diversity rather than gene expansion for functional regulation. Overall, the comparative analysis confirms that the motif architecture of *S. spontaneum* represents the ancestral blueprint of the AP2/ERF superfamily in sugarcane.

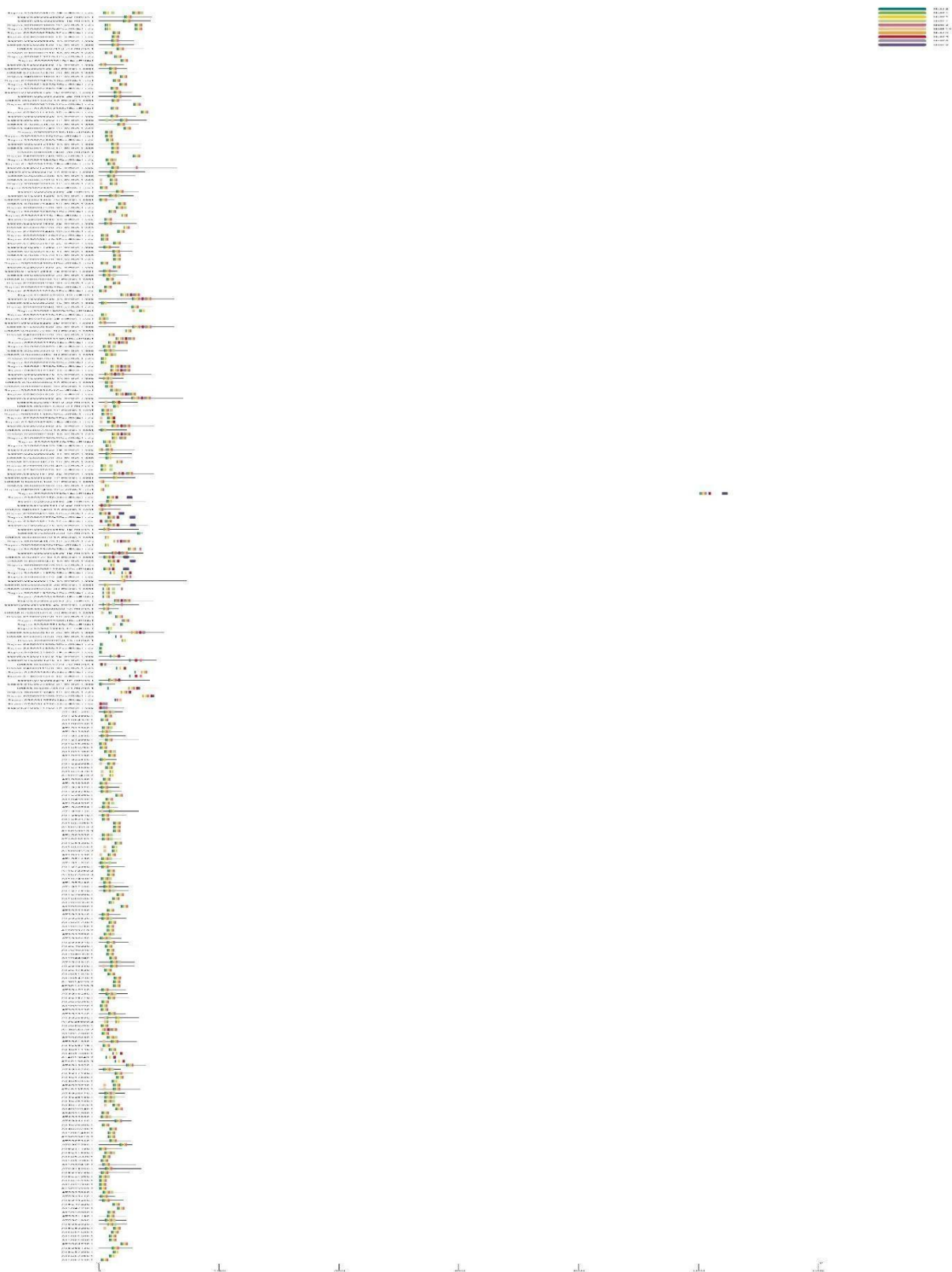


Figure 5.12: Conserved motif distribution of AP2/ERF proteins in the merged dataset of *S. spontaneum* and *Arabidopsis thaliana*, as identified by MEME and visualized using TBtools

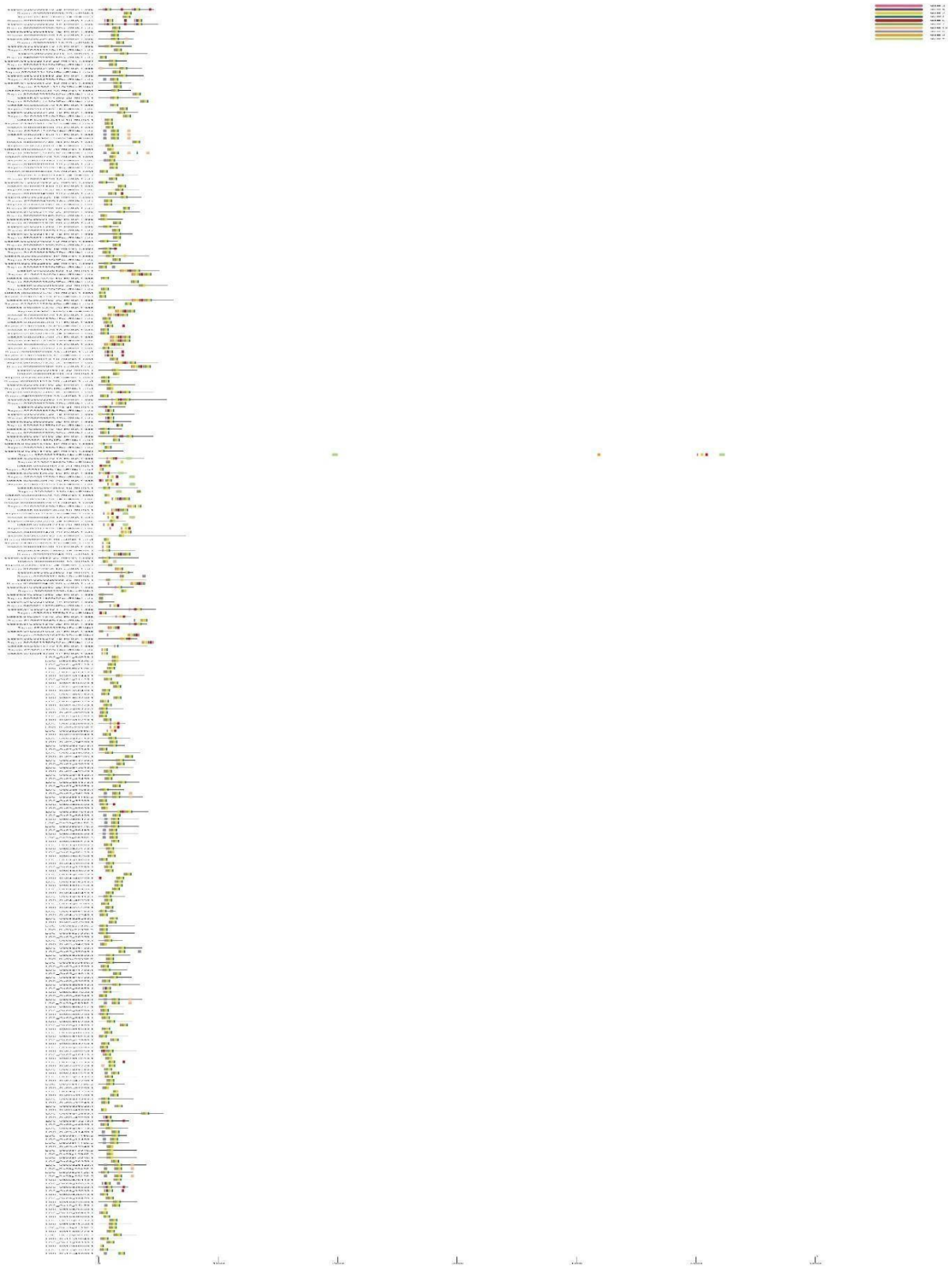


Figure 5.13: Conserved motif distribution of AP2/ERF proteins in the merged dataset of *S. spontaneum* and *Oryza sativa*, as identified by MEME and visualized using TBtools

5.4 Phylogenetic Analysis

Phylogenetic analysis was performed to examine the evolutionary relationships among ERF gene family members across the studied datasets. The aligned protein sequences were used to construct a Maximum Likelihood (ML) tree based on the JTT substitution model in MEGA 12, with 500 bootstrap replicates to assess the reliability of the topology.

The resulting tree showed clear clustering of ERF proteins into distinct clades, with moderate to high bootstrap support. Genes from sugarcane and reference species were distributed across multiple clades, indicating a conserved evolutionary origin along with species-specific diversification. For systematic classification, the clades were assigned to subgroups based on the previously reported ERF classification in *Arabidopsis thaliana*, in which members are categorised into Groups I to X. In addition, *Oryza sativa* ERF proteins were classified into I to XIV. These subgroup classifications were adopted to distinguish and annotate the clades in the present study, and were represented using Roman numerals for consistency.

Visualization using iTOL enabled clear identification of clustering patterns and subgroup distribution across datasets.

5.4.1 Phylogenetic Classification of *S. officinarum* Using *Arabidopsis thaliana*

To investigate the evolutionary relationships of the ERF gene family, we constructed a phylogenetic tree using a combined dataset of 1,336 *S. officinarum* sequences and 139 *Arabidopsis thaliana* sequences. Based on the established *Arabidopsis* ERF classification framework, the sequences were assigned into ten major subgroups (I–X). In addition, distinct lineage-associated clusters were identified and designated as VI-L and Xb-L.

Our analysis revealed that the majority of *S. officinarum* genes were distributed across these defined subgroups, indicating a conserved phylogenetic framework between sugarcane and *Arabidopsis*. However, the distribution of genes across subgroups was uneven. Group VII comprised the largest cluster (255 genes), followed by Group III (172 genes) and Group IX (153 genes), suggesting subgroup-specific expansion within the ERF family. The DREB subfamily (Groups I–IV) was well represented, with 66, 88, 172, and 58 genes respectively, while the ERF subfamily (Groups V–X) also exhibited variable representation, particularly in Groups VIII (135 genes) and Xa (102 genes). The lineage-associated subgroups VI-L (17 genes) and Xb-L (15 genes) were comparatively smaller but consistently resolved as distinct clades.

In addition to the major subgroups, a small set of sequences did not cluster within the established classification. These were designated as Soloist (SOR), comprising 3 genes, and represent phylogenetically distinct members of the ERF family. Furthermore, 182 genes were categorized as unclassified (UN), as they did not align with any defined subgroup, indicating potential lineage-specific divergence or structural variation within *S. officinarum*.

Overall, the phylogenetic analysis demonstrates that the ERF gene family in sugarcane largely conforms to the conserved *Arabidopsis*-based classification, while exhibiting subgroup-specific expansion and the presence of distinct unclassified members, reflecting the evolutionary complexity of this gene family.

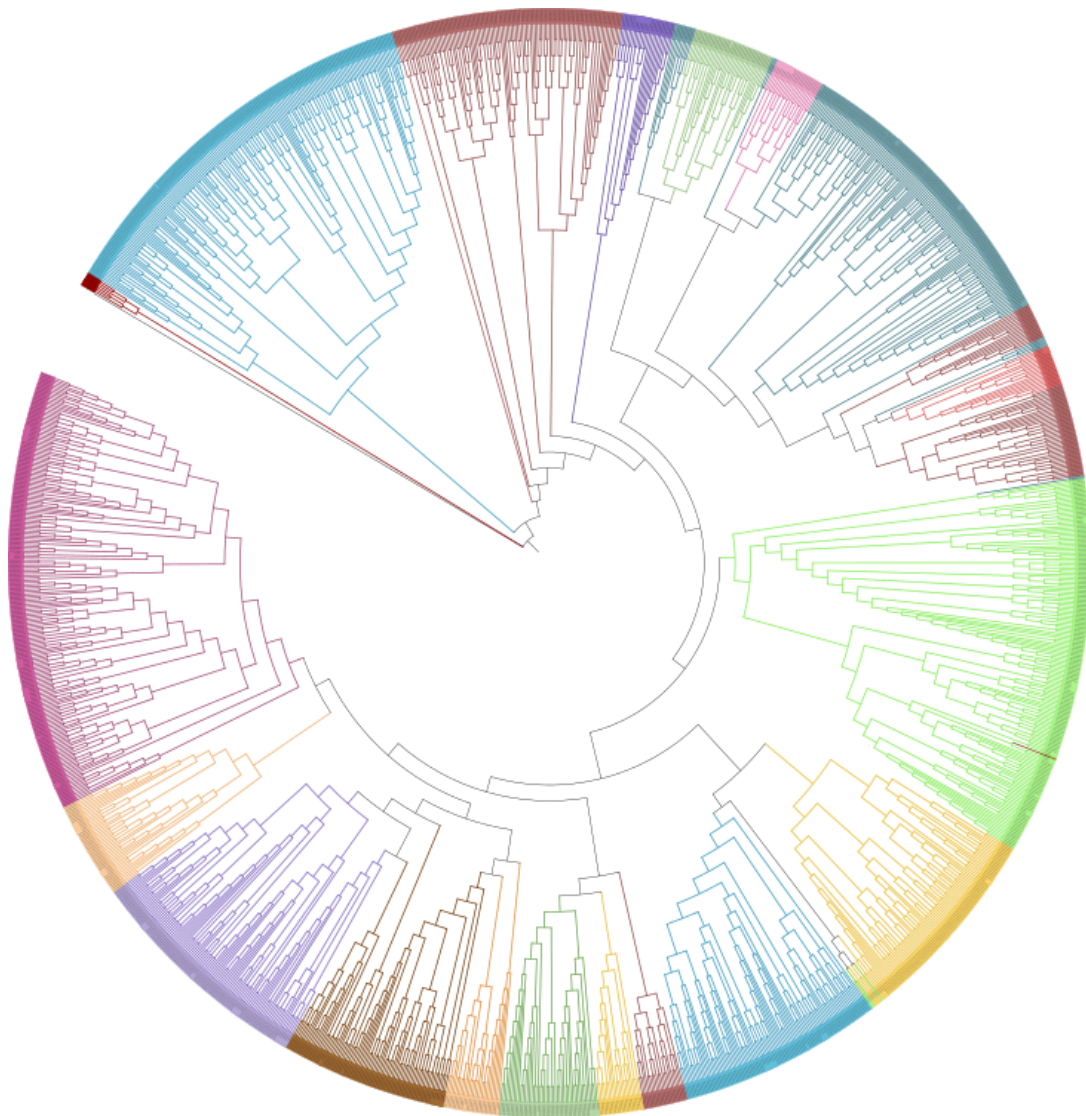


Figure 5.14: Maximum Likelihood phylogenetic tree of ERF proteins from *S. officinarum* and *Arabidopsis thaliana*, showing classification into subgroups I–X, VI–L, Xb–L, SOR, and UN.

Table 5.2 Subgroup Distribution of *S. officinarum* ERF Genes (*Arabidopsis*-Reference Tree)

Subgroup	Gene Count
I	66
II	88
III	172
IV	58
V	76
VI	14
VII	255
VIII	135
IX	153
Xa	102
VI-L	17
Xb-L	15
SOR	3
UN	182

5.4.2 Phylogenetic Classification of *S. officinarum* Using *Oryza sativa*

The phylogenetic tree constructed using *S. officinarum* (La purple) ERF sequences in combination with *Oryza sativa* reference sequences enabled classification based on the *Oryza sativa*-specific ERF framework (Groups I–XIV). In addition to the primary groups, a distinct lineage-associated cluster (VI-L) and an unclassified (UN) category were also identified.

From our analysis, *S. officinarum* genes were distributed across most of the defined subgroups, although the distribution was uneven. Group IX comprised the largest cluster (229 genes), followed by Group VIII (151 genes), indicating notable expansion of these subgroups in

sugarcane. In addition, Group III (144 genes) and Group VII (143 genes) were also highly represented, suggesting their relative enrichment within the gene family.

The remaining subgroups also showed variable representation, including Group I (72), Group II (69), Group IV (52), Group V (73), Group VI (31), and Group X (104). The lineage-associated subgroup VI-L (21 genes) formed a smaller but clearly distinct cluster.

Interestingly, no genes were assigned to Groups XI, XII, and XIV, indicating their absence or possible divergence in *S. officinarum*. In addition, 102 genes were categorized as unclassified (UN), as they did not cluster within the defined subgroups, suggesting lineage-specific diversification within the sugarcane ERF gene family.

Overall, the phylogenetic classification based on the *Oryza sativa* reference system indicates that while *S. officinarum* follows a conserved monocot-specific framework, certain subgroups show notable expansion along with the presence of unclassified members.

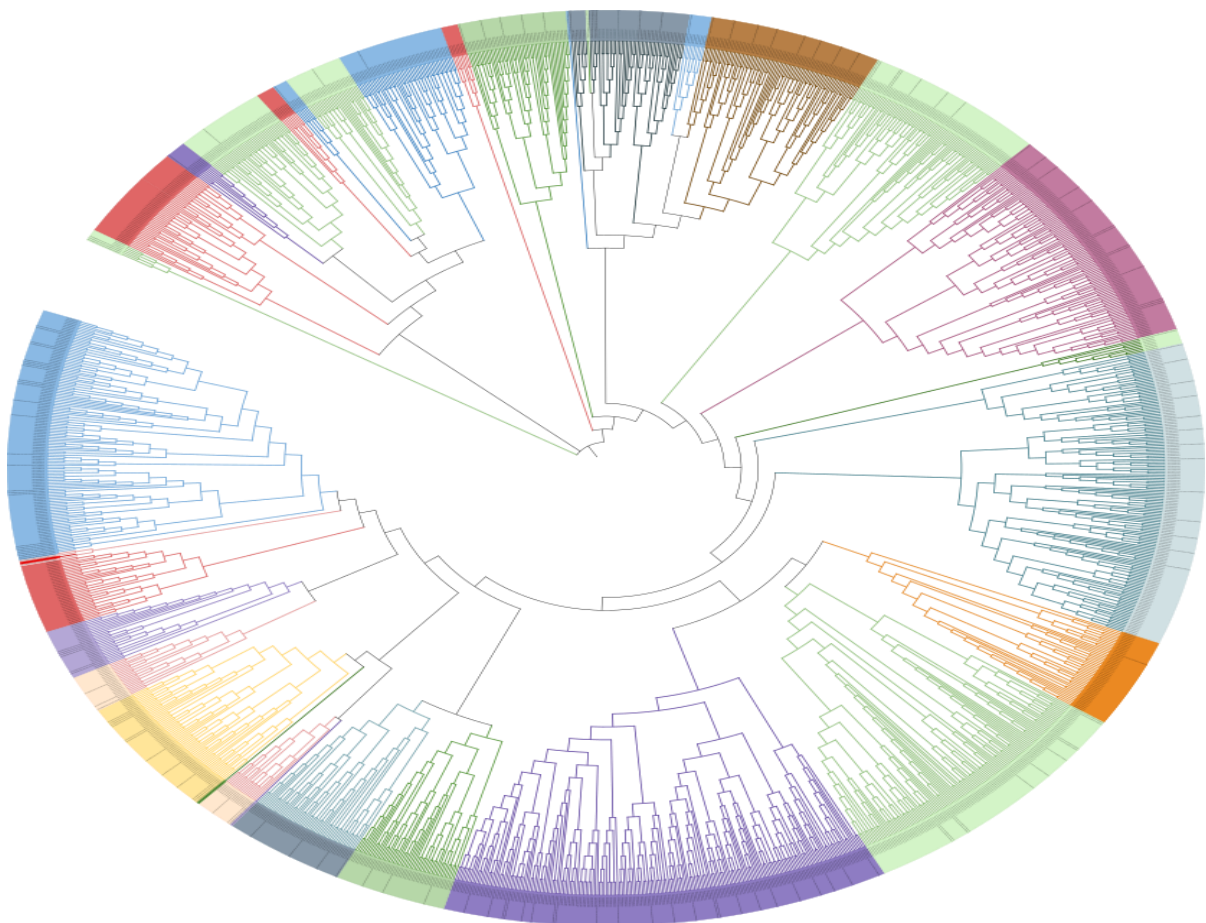


Figure 5.15: Maximum Likelihood phylogenetic tree of ERF proteins from *S. officinarum* and *Oryza sativa*, showing classification into subgroups I–XIV, VI-L, and UN.

Table 5.3: Subgroup Distribution of *S. officinarum* ERF Genes (*Oryza sativa*-Reference Tree)

Subgroup	Gene Count
I	72
II	69
III	144
IV	52
V	73
VI	31
VII	143
VIII	151
IX	229
X	104
XI	0
XII	0
XIV	0
VI-L	21
UN	102

5.4.3 Phylogenetic Classification of R570 Hybrid Using *Arabidopsis thaliana*

The phylogenetic tree constructed using R570 hybrid ERF sequences in combination with *Arabidopsis thaliana* reference sequences enabled classification based on the established *Arabidopsis* ERF framework (subgroups I–X). In addition to the primary subgroups, distinct lineage-associated clusters (VI-L and Xb-L) as well as Soloist (SOR) and unclassified (UN) categories were identified.

From our analysis, R570 genes were distributed across all major subgroups, although the distribution was uneven. Subgroup VII comprised the largest cluster (195 genes), followed by Subgroup III (132 genes) and Subgroup IX (117 genes), indicating notable expansion of specific subgroups in the hybrid genome. Several other subgroups also showed moderate representation, including Subgroup VIII (103 genes) and Xa (78 genes), while Subgroups I (51), II (68), IV (45), V (58), and VI (11) contained comparatively fewer genes. The lineage-associated subgroups VI-L (13 genes) and Xb-L (11 genes) formed smaller but clearly distinguishable clusters.

In addition, a small number of sequences were classified under the Soloist (SOR) category (2 genes), representing conserved and distinct members of the ERF family. A substantial proportion of genes (136) were categorized as unclassified (UN), as they did not cluster within the defined subgroups, suggesting divergence and possible lineage-specific variation within the R570 hybrid.

Overall, the phylogenetic distribution indicates that while the R570 hybrid largely follows the conserved *Arabidopsis*-based classification, certain subgroups exhibit expansion along with the presence of unclassified members, reflecting the genomic complexity of the hybrid

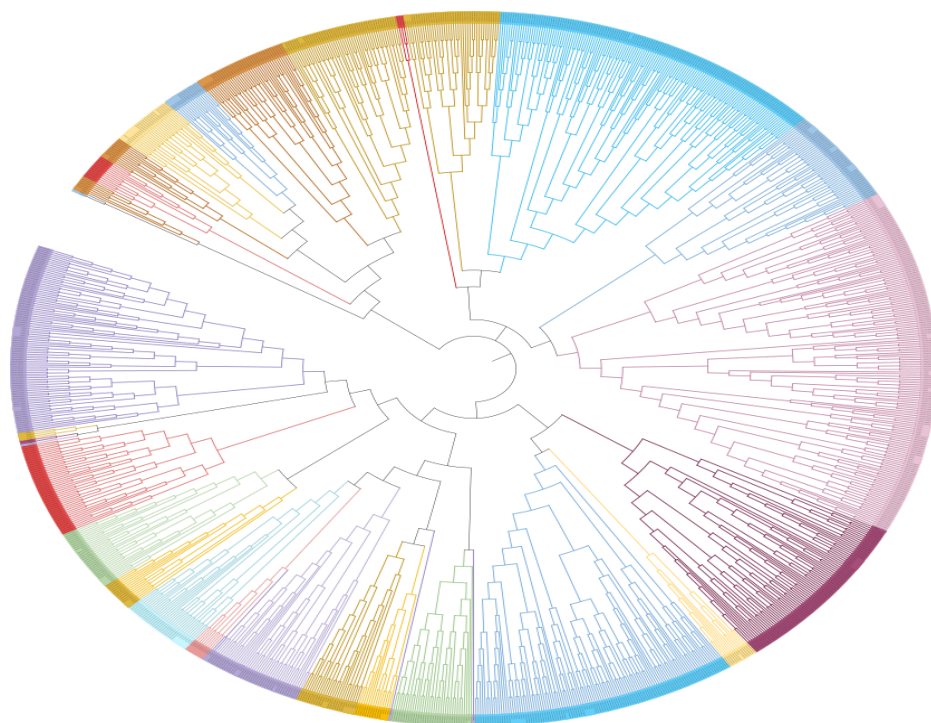


Figure 5.16: Maximum Likelihood phylogenetic tree of ERF proteins from R570 hybrid and *Arabidopsis thaliana*, showing classification into subgroups I–X, VI-L, Xb-L, SOR, and UN.

Table 5.4: Subgroup Distribution of R570 Hybrid ERF Genes (*Arabidopsis*-Reference Tree)

Subgroup	Gene Count
I	51
II	68
III	132
IV	45
V	58
VI	11
VII	195
VIII	103
IX	117
Xa	78
VI-L	13
Xb-L	11
SOR	2
UN	136

5.4.4 Phylogenetic Classification of R570 Hybrid Using *Oryza sativa*

The phylogenetic tree constructed using R570 hybrid ERF sequences in combination with *Oryza sativa* reference sequences enabled classification based on the *Oryza sativa*-specific ERF framework (subgroups I–XIV). In addition to the primary subgroups, a lineage-associated cluster (VI-L) and an unclassified (UN) category were also identified.

From our analysis, R570 genes were distributed across most of the defined subgroups, although the distribution was uneven. Subgroup IX comprised the largest cluster (178 genes), followed by Subgroup VIII (121 genes), indicating notable expansion of these subgroups in the hybrid

genome. In addition, Subgroup VII (112 genes) and Subgroup III (104 genes) were also highly represented, suggesting their relative enrichment within the ERF gene family.

The remaining subgroups showed moderate to lower representation, including Subgroup I (52), Subgroup II (48), Subgroup IV (38), Subgroup V (51), Subgroup VI (23), and Subgroup X (81). The lineage-associated subgroup VI-L (16 genes) formed a smaller but distinct cluster.

Interestingly, no genes were assigned to Subgroups XI, XII, and XIV, indicating their absence or possible divergence in the R570 hybrid. In addition, a substantial number of genes (196) were categorized as unclassified (UN), as they did not cluster within the defined subgroups, suggesting divergence and lineage-specific variation within the hybrid genome.

Overall, the phylogenetic classification based on the *Oryza sativa* reference system indicates that while the R570 hybrid follows a conserved monocot-specific framework, certain subgroups show notable expansion along with a high proportion of unclassified members, reflecting the genomic complexity of the hybrid.

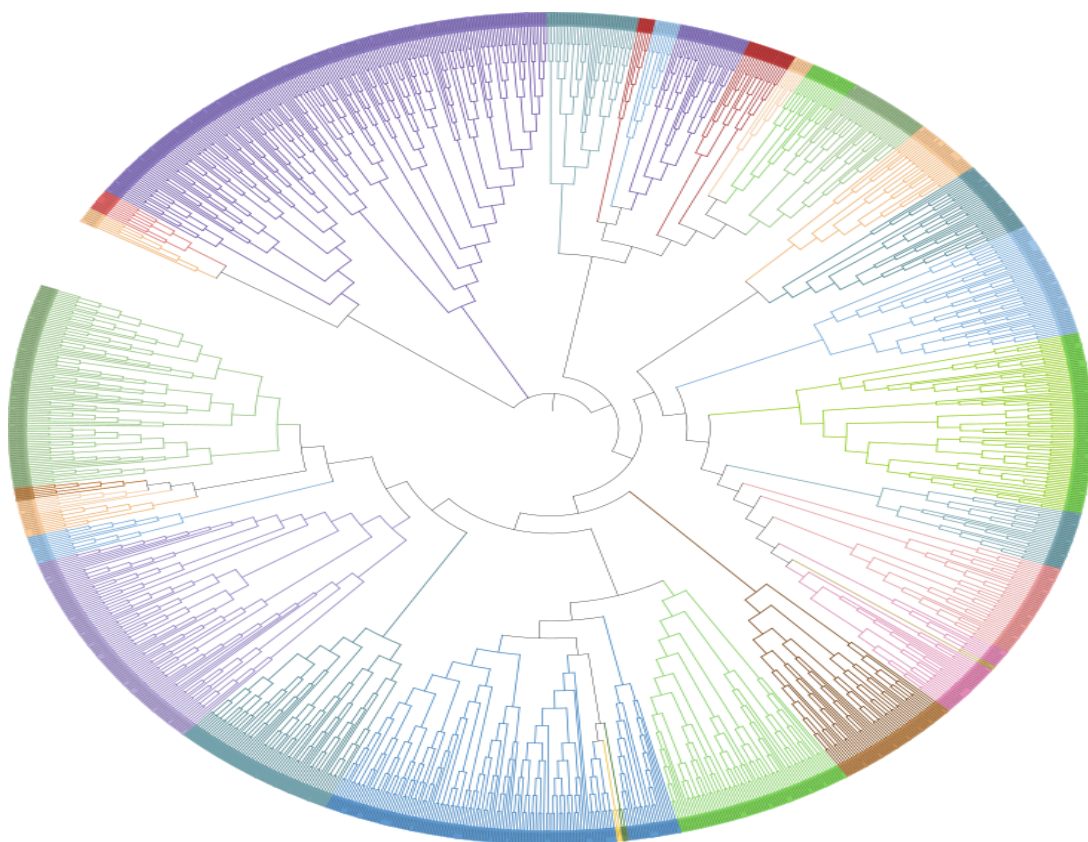


Figure 5.17: Maximum Likelihood phylogenetic tree of ERF proteins from R570 hybrid and *Oryza sativa*, showing classification into subgroups I–XIV, VI-L, and UN.

Table 5.5: Subgroup Distribution of R570 Hybrid ERF Genes (*Oryza sativa*-Reference Tree)

Subgroup	Gene Count
I	52
II	48
III	104
IV	38
V	51
VI	23
VII	112
VIII	121
IX	178
X	81
XI	0
XII	0
XIV	0
VI-L	16
UN	196

5.4.5 Phylogenetic Classification of *S. spontaneum* Using *Arabidopsis thaliana*

The phylogenetic tree constructed using *S. spontaneum* ERF sequences in combination with *Arabidopsis thaliana* reference sequences enabled classification based on the established *Arabidopsis* ERF framework (subgroups I–X). In addition to the primary subgroups, lineage-associated clusters (VI-L and Xb-L) as well as Soloist (SOR) and unclassified (UN) categories were identified.

From our analysis, genes of *S. spontaneum* were distributed across all major subgroups, although the overall gene count was comparatively lower. Subgroup VII comprised the largest cluster (28 genes), followed by Subgroup III (24 genes) and Subgroup IX (21 genes), indicating their relative enrichment even in the ancestral genome. Other subgroups showed moderate representation, including Subgroup VIII (18 genes), Xa (13 genes), II (14 genes), I (12 genes), V (11 genes), and IV (10 genes), while Subgroup VI (4 genes) contained the fewest members. The lineage-associated subgroups VI-L (3 genes) and Xb-L (2 genes) formed small but distinct clusters.

In addition, a single gene was classified under the Soloist (SOR) category, representing a conserved member of the ERF family. A total of 15 genes were categorized as unclassified (UN), as they did not cluster within the defined subgroups, indicating early divergence within the *Saccharum* lineage.

Overall, the phylogenetic distribution suggests that *S. spontaneum* possesses a conserved yet relatively compact ERF gene repertoire, which likely represents the ancestral framework from which the expanded gene families of cultivated and hybrid sugarcane have evolved.

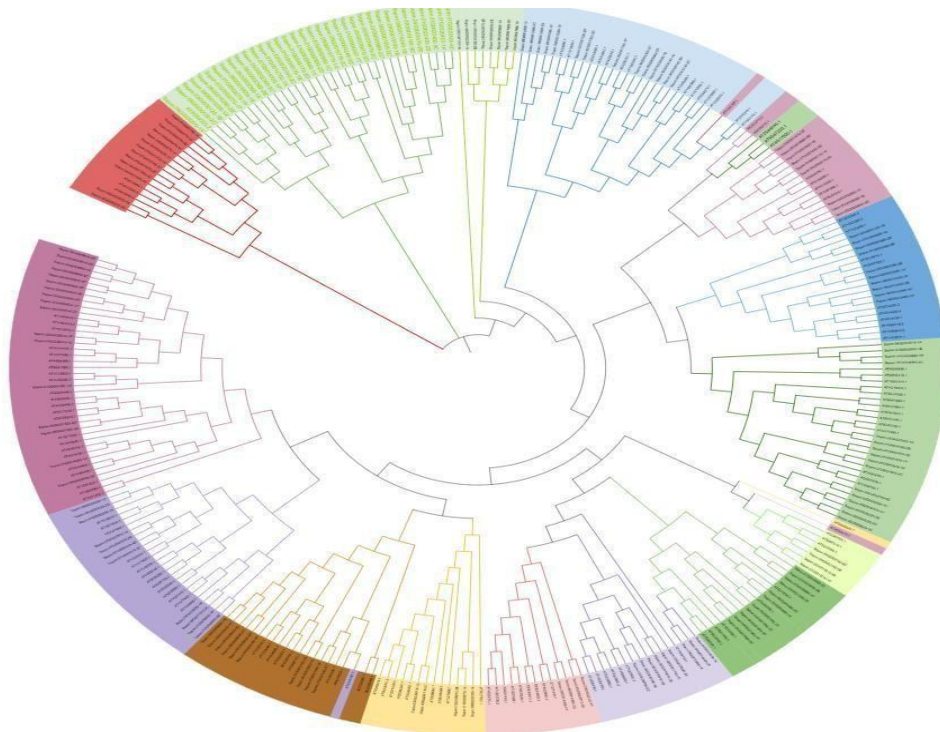


Figure 5.18: Maximum Likelihood phylogenetic tree of ERF proteins from *S. spontaneum* and *Arabidopsis thaliana*, showing classification into subgroups I–X, VI-L, Xb-L, SOR, and UN.

Table 5.6: Subgroup Distribution of *S. spontaneum* ERF Genes (*Arabidopsis*-Reference Tree)

Subgroup	Gene Count
I	12
II	14
III	24
IV	10
V	11
VI	4
VII	28
VIII	18
IX	21
Xa	13
VI-L	3
Xb-L	2
SOR	1
UN	15

5.4.6 Phylogenetic Classification of *S. spontaneum* Using *Oryza sativa*

The phylogenetic tree constructed using *S. spontaneum* ERF sequences in combination with *Oryza sativa* reference sequences enabled classification based on the *Oryza sativa*-specific ERF framework (subgroups I–XIV). In addition to the primary subgroups, a lineage-associated cluster (VI-L) and an unclassified (UN) category were identified.

From our analysis, genes of *S. spontaneum* were distributed across most of the defined subgroups, although the overall gene count remained comparatively low. Subgroup VII comprised the largest cluster (28 genes), followed by Subgroup III (24 genes) and Subgroup

IX (21 genes), indicating their relative enrichment within the ancestral genome. Other subgroups showed moderate representation, including Subgroup VIII (18 genes), Subgroup X (13 genes), Subgroup II (14 genes), Subgroup I (12 genes), Subgroup V (11 genes), and Subgroup IV (10 genes), while Subgroup VI (5 genes) contained fewer members. The lineage-associated subgroup VI-L (3 genes) formed a small but distinct cluster.

Interestingly, no genes were assigned to Subgroups XI, XII, and XIV, indicating their absence or possible early divergence in *S. spontaneum*. In addition, 17 genes were categorized as unclassified (UN), as they did not cluster within the defined subgroups, suggesting early lineage-specific variation within the *Saccharum* genus.

Overall, the phylogenetic classification based on the *Oryza sativa* reference system indicates that *S. spontaneum* retains a conserved and compact ERF gene repertoire, representing the ancestral framework from which the expanded gene families of cultivated and hybrid sugarcane have evolved.

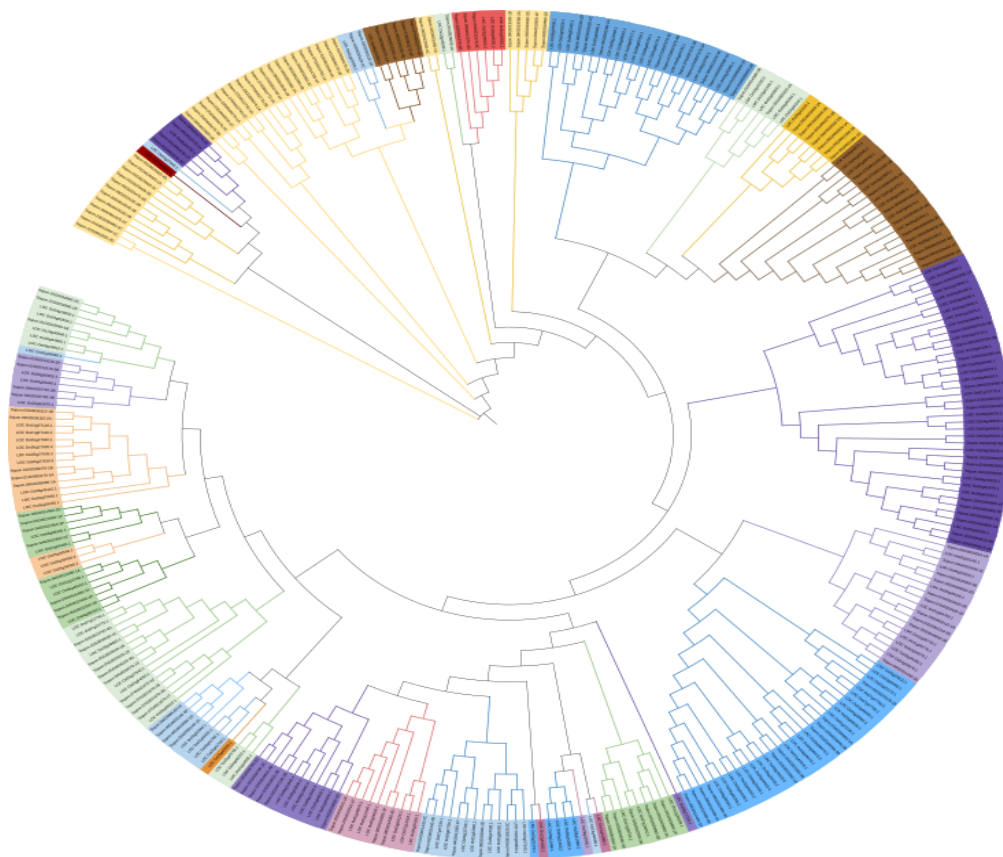


Figure 5.19: Maximum Likelihood phylogenetic tree of ERF proteins from *S. spontaneum* and *Oryza sativa*, showing classification into subgroups I–XIV, VI–L, and UN.

Table 5.7: Subgroup Distribution of *S. spontaneum* ERF Genes (*Oryza sativa*-Reference Tree)

Subgroup	Gene Count
I	12
II	14
III	24
IV	10
V	11
VI	5
VII	28
VIII	18
IX	21
X	13
XI	0
XII	0
XIV	0
VI-L	3
UN	17

5.5 Intron–Exon Organization of AP2/ERF Genes

The exon–intron organization of AP2/ERF genes was analyzed to investigate the structural architecture of the gene family across sugarcane genomes. For this analysis, only sugarcane genome datasets (*S. officinarum*, R570, and *S. spontaneum*) were used, and the merged dataset was not included to ensure genome-specific accuracy.

Gene structures were visualized using TBtools by aligning coding sequences (CDS) with their corresponding genomic sequences. Two input files were provided for visualization: (i) a gene position file containing gene IDs along with chromosome number, start, and end positions, and (ii) a chromosome length file specifying the size of each chromosome.

The graphical representation included coding sequences (CDS), untranslated regions (UTRs), and introns, which were distinguished based on color coding. CDS regions were represented as blocks, UTRs as flanking regions, and introns as connecting lines between exons.

5.5.1 *S. officinarum*

The gene structure analysis of AP2/ERF genes in *S. officinarum* revealed a diverse range of exon–intron organizations. Gene structure visualization showed that coding sequences (CDS) are represented by green blocks, untranslated regions (UTRs) by yellow blocks, and introns by thin black lines.

A predominant pattern observed was the presence of simple gene structures, where many genes consisted of continuous green CDS blocks with little to no interruption by black intronic lines, indicating an intronless or intron-poor architecture. In addition to these compact structures, a subset of genes exhibited more complex exon–intron arrangements, characterized by multiple green CDS blocks separated by introns. These genes showed a conserved organization pattern, suggesting structural stability within specific subfamilies. The yellow UTR regions displayed noticeable variation in length and distribution among different genes, indicating potential differences in regulatory elements associated with transcriptional and post-transcriptional control.

Overall, the gene structure patterns observed in *S. officinarum* reflect a combination of structural conservation and variation, highlighting the complexity of the AP2/ERF gene family in cultivated sugarcane.



Figure 5.20: Exon–intron organization of AP2/ERF genes in *S. officinarum* visualized using TBtools. Coding sequences (CDS) are represented by green blocks, untranslated regions (UTRs) by yellow blocks, and introns by thin black lines.

5.5.2 R570 Hybrid

The exon–intron organization of AP2/ERF genes in the R570 hybrid was analyzed to understand the structural diversity within this gene family. Gene structure visualization revealed variations in the arrangement of coding and non-coding regions across the identified genes, where coding sequences (CDS) are represented by green blocks, untranslated regions (UTRs) by yellow blocks, and introns by thin black lines.

A predominant pattern observed was the presence of genes with a single continuous green CDS block, indicating an intronless architecture. This structural simplicity is commonly associated with members of specific AP2/ERF subfamilies and suggests a conserved gene organization within the hybrid genome. The yellow UTR regions exhibited variability in length and distribution among different genes, indicating potential differences in regulatory elements associated with gene expression and post-transcriptional control.

In contrast, a subset of genes displayed multiple green CDS blocks separated by thin black lines (introns), reflecting a more complex exon–intron organization. These genes are typically associated with subfamilies known to possess conserved multi-exon structures.

Overall, the gene structure analysis highlights both conserved and variable architectural features within the AP2/ERF gene family in the R570 hybrid.



Figure 5.21: Exon–intron organization of AP2/ERF genes in R570 visualized using TBtools. Coding sequences (CDS) are represented by green blocks, untranslated regions (UTRs) by yellow blocks, and introns by thin black lines

5.5.3 *S. spontaneum*

The gene structure analysis of AP2/ERF genes in *S. spontaneum* was performed to understand the structural organization of this gene family in the wild sugarcane genome. Gene structure visualization indicated that coding sequences (CDS) are represented by green blocks, untranslated regions (UTRs) by yellow blocks, and introns by thin black lines.

A predominant pattern observed was the presence of genes with simple structures, where many genes consisted of a single continuous green CDS block with little to no interruption by introns, indicating an intronless or intron-poor organization. In addition to these simple structures, a subset of genes exhibited more complex exon–intron arrangements, characterized by multiple green CDS blocks separated by thin black lines. These structures were consistently observed in specific subfamilies, indicating conserved gene organization. The yellow UTR regions showed variation in length and distribution, suggesting potential differences in regulatory elements associated with gene expression.

Overall, the gene structure patterns observed in *S. spontaneum* indicate both conserved and variable structural features within the AP2/ERF gene family.

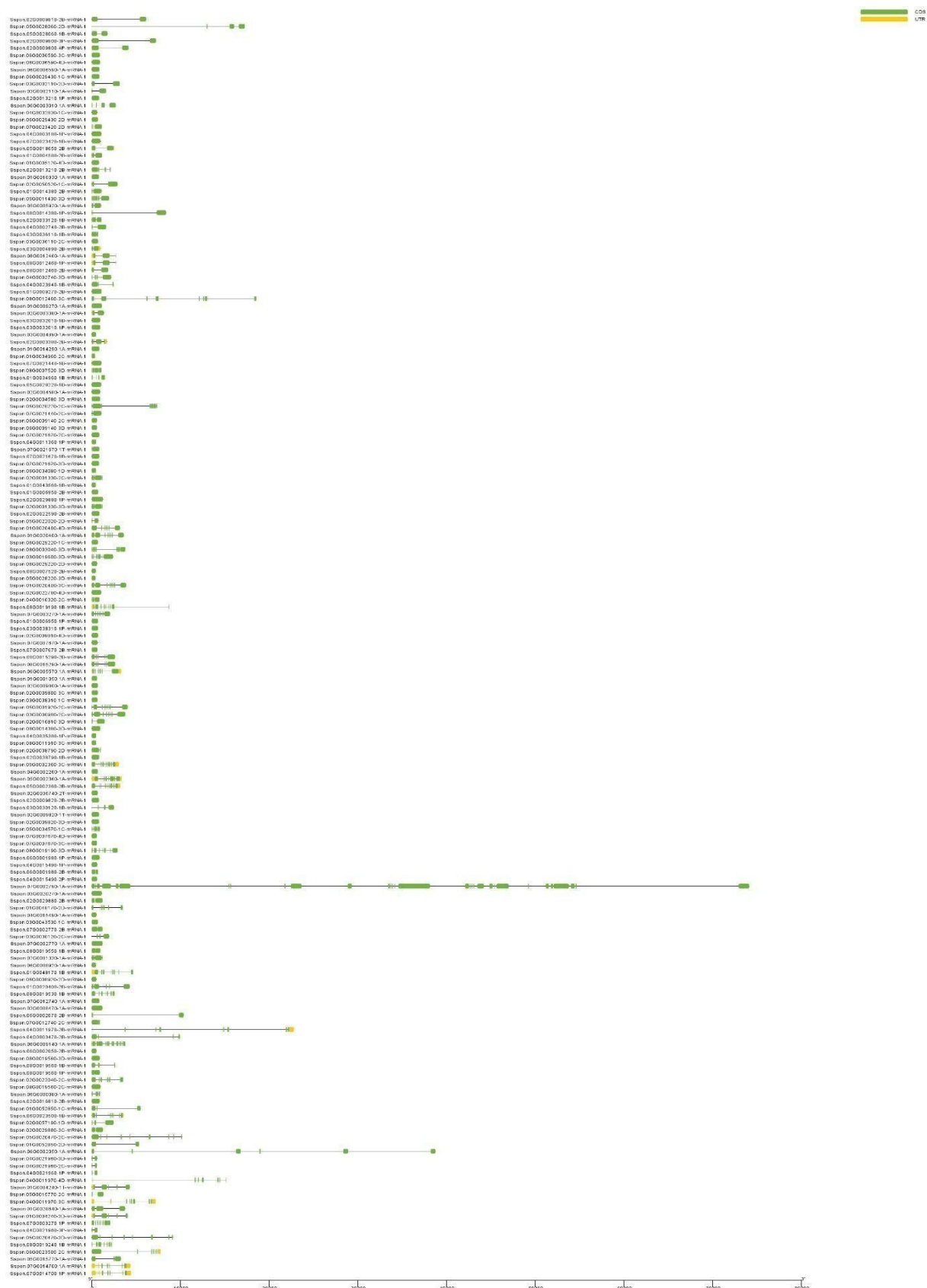


Figure 5.22: Exon–intron organization of AP2/ERF genes in *S. spontaneum* visualized using TBtools. Coding sequences (CDS) are represented by green blocks, untranslated regions (UTRs) by yellow blocks, and introns by thin black lines.

5.6 Chromosomal Localization of AP2/ERF Genes

Chromosomal localization of AP2/ERF genes was performed to determine their physical distribution across sugarcane genomes. The analysis was carried out using TBtools, where genomic coordinates were used to map genes onto their respective chromosomes.

For this analysis, only sugarcane genome datasets were used, and the merged dataset was not included to maintain genome-specific accuracy. Two input files were required for visualization: (i) a gene position file containing gene IDs along with chromosome number, start, and end positions, and (ii) a chromosome length file specifying the size of each chromosome.

The genes were mapped onto chromosomal scaffolds using the Genome Location Mapper tool in TBtools to visualize their distribution patterns.

5.6.1 *S. officinarum*

The chromosomal mapping of AP2/ERF genes in *S. officinarum* (LA Purple) revealed that the genes are distributed across the genome and are spread over 10 basic chromosomes (Chr01–Chr10). Each chromosome is represented by multiple homologous copies, typically designated as A to H, reflecting the polyploid nature of the genome. For consistency and ease of interpretation, all genes were systematically renamed as *LAERF1*, *LAERF2*, *LAERF3*, and so on.

The distribution pattern indicates that AP2/ERF genes are widely dispersed across these chromosomes rather than being confined to specific regions. However, the number of genes varies among chromosomes, with some chromosomes showing relatively higher gene density, while others contain fewer genes, indicating an uneven distribution pattern.

In addition to the dispersed distribution, certain chromosomal regions exhibit localized clustering of genes, suggesting the presence of gene-rich areas within the genome.

Overall, the chromosomal localization analysis highlights a widespread and non-uniform distribution of AP2/ERF genes across the *S. officinarum* genome.



Figure 5.23: Chromosomal localization of AP2/ERF genes in *S. officinarum*, showing distribution across chromosomes Chr01–Chr10 (A–H)

5.6.2 R570 Hybrid

The chromosomal mapping of AP2/ERF genes in the R570 hybrid was performed to analyze their distribution across the genome. The identified genes were distributed across all ten basic chromosome groups (Chr01–Chr10), with each chromosome represented by multiple homologous sets (A–F), reflecting the polyploid nature of the hybrid genome. For consistency and ease of interpretation, all genes were systematically renamed as *R570ERF1*, *R570ERF2*, *R570ERF3*, and so on.

The distribution pattern indicates that AP2/ERF genes are widely dispersed across the chromosomes rather than being confined to specific regions. However, variation in gene density was observed among different chromosomes, with certain chromosomes containing relatively higher numbers of genes, while others showed fewer genes, indicating an uneven distribution pattern.

In addition to the dispersed distribution, some chromosomal regions exhibited clustering of genes, where multiple AP2/ERF loci were located in close proximity. These clustered regions may represent gene-rich areas within the genome and suggest possible expansion of the gene family in specific chromosomal regions.

Overall, the chromosomal localization analysis highlights a widespread and non-uniform distribution of AP2/ERF genes across the R570 hybrid genome.



Figure 5.24: Chromosomal localization of AP2/ERF genes in the R570 hybrid, showing distribution across chromosomes Chr01–Chr10 (A–F).

5.6.3 *S. spontaneum*

The chromosomal mapping of AP2/ERF genes in *S. spontaneum* was performed to analyze their distribution across the genome. A total of 176 AP2/ERF genes were identified and distributed across eight basic chromosome groups (Chr1–Chr8), with each chromosome represented by four homologous sets (A–D), reflecting the polyploid nature of the genome. For consistency and ease of interpretation, all identified genes were renamed as *SPONERF1*, *SPONERF2*, *SPONERF3*, and so on.

The distribution pattern indicates that AP2/ERF genes are widely dispersed across all chromosomes rather than being confined to specific regions. However, variation in gene density was observed among different chromosomes, with certain chromosomes containing relatively higher numbers of genes, while others showed fewer genes, indicating an uneven distribution pattern.

In addition to the dispersed distribution, some chromosomal regions exhibited clustering of genes, where multiple AP2/ERF loci were located in close proximity. These clustered regions may represent gene-rich areas within the genome and suggest possible expansion of the gene family in specific chromosomal regions.

Overall, the chromosomal localization analysis highlights a widespread and non-uniform distribution of AP2/ERF genes across the *S. spontaneum* genome.



Figure 5.25: Chromosomal localization of AP2/ERF genes in *S. spontaneum*, showing distribution across chromosomes Chr1–Chr8 (A–D).

5.7 Physicochemical Properties and Subcellular Localization of AP2/ERF Genes

The physicochemical properties and subcellular localization of AP2/ERF proteins were analyzed using a combined dataset comprising sequences from *Arabidopsis thaliana*, *Oryza sativa*, and three sugarcane genomes (*S. spontaneum*, *S. officinarum*, and the R570 hybrid). This integrated analysis was performed to provide a comparative overview of structural characteristics and functional attributes across model plants and sugarcane species.

The results revealed considerable variation in protein length, molecular weight, and isoelectric point (pI), indicating structural diversity within the AP2/ERF gene family. Despite this variability, a consistent pattern of predominant nuclear localization was observed across all species, supporting the conserved role of these proteins as transcription factors. Minor variations in localisation patterns and physicochemical properties among species suggest functional diversification within the gene family.

5.7.1 *S. spontaneum*

The physicochemical properties and subcellular localization of AP2/ERF proteins were analyzed using combined datasets comprising reference and sugarcane species. Key parameters including protein length, molecular weight (MW), isoelectric point (pI), and predicted subcellular localization were evaluated.

The combined dataset of *Arabidopsis thaliana* and *S. officinarum* revealed a wide variation in protein length, ranging from 103 amino acids (*Soffic.04G0009570-4G.t1*) to 1780 amino acids (*Soffic.08G0006230-8D.t1*). The molecular weight ranged from 11,568.7 Da to 199,344.33 Da, reflecting differences in protein size and composition. The isoelectric point (pI) values varied between 4.1 and 11.83, indicating the presence of both acidic and basic proteins. Subcellular localization analysis showed that the majority of proteins were localized in the nucleus (1233 proteins), followed by the chloroplast (155 proteins), cytoplasm (38 proteins), and mitochondria (37 proteins). Very few proteins were localized in the extracellular region (8 proteins) and plasma membrane (4 proteins).

Similarly, the combined dataset of *S. officinarum* and *Oryza sativa* also exhibited considerable variation in physicochemical properties. The protein length ranged from 103 amino acids (*Soffic.04G0009570-4G.t1*) to 1780 amino acids (*Soffic.08G0006230-8D.t1*), while the molecular weight varied from 11,568.7 Da to 199,344.33 Da. The isoelectric point (pI) ranged from 4.1 to 11.83, indicating a similar distribution of acidic and basic proteins. Subcellular

localization analysis revealed that most proteins were predominantly localized in the nucleus (1234 proteins), followed by the chloroplast (170 proteins), cytoplasm (44 proteins), and mitochondria (36 proteins). A very small number of proteins were observed in the extracellular region (9 proteins) and plasma membrane (5 proteins).

Overall, the combined analysis indicates that AP2/ERF proteins exhibit significant variation in physicochemical properties while maintaining a predominantly nuclear localization pattern across datasets, supporting their conserved role as transcription factors.

Table 5.8: Physicochemical Properties of *S. officinarum*

Dataset	Length (aa)	MW (Da)	pI
<i>A. thaliana</i> + <i>S. officinarum</i>	103–1780	11,568.7–199,344.33	4.1–11.83
<i>O. sativa</i> + <i>S. officinarum</i>	103–1780	11,568.7–199,344.33	4.1–11.83

Table 5.9: Subcellular Localization of *S. officinarum*

Dataset	Nuclear	Chloroplast	Cytoplasm	Mitochondria	Extracellular	Plasma Membrane
<i>A. thaliana</i> + <i>S. officinarum</i>	1233	155	38	37	8	4
<i>O. sativa</i> + <i>S. officinarum</i>	1234	170	44	36	9	5

5.7.2 R570

The physicochemical properties of ERF proteins identified from the combined dataset of R570 and *Arabidopsis thaliana* showed considerable variation in protein length, molecular weight, and isoelectric point. The protein length ranged from 107 amino acids (*SoffiXsponR570.03Bg045800.1*) to 704 amino acids (*SoffiXsponR570.02Fg335900.1*). The

molecular weight varied between 11,313.69 Da and 73,607.54 Da, corresponding to the same minimum and maximum length proteins, respectively.

The isoelectric point (pI) ranged from 4.42 (*SoffiXsponR570.04Ag090900.1*) to 12, with multiple proteins exhibiting the highest pI values. Subcellular localization prediction indicated that the majority of proteins were localized in the nucleus (977 proteins), followed by chloroplast (135 proteins), cytoplasm (26 proteins), mitochondria (16 proteins), and extracellular regions (5 proteins).

Similarly, in the combined dataset of R570 and *Oryza sativa*, ERF proteins exhibited a wide range of physicochemical properties. The protein length ranged from 107 amino acids (*SoffiXsponR570.03Bg045800.1*) to 704 amino acids (*SoffiXsponR570.02Fg335900.1*), consistent with the previous dataset. The molecular weight ranged from 11,313.69 Da to 73,607.54 Da.

The isoelectric point (pI) ranged from 4.28 (*LOC_Os05g27930.2*) to 12, indicating the presence of both acidic and basic proteins. Subcellular localization analysis showed that most proteins were predicted to be localized in the nucleus (978 proteins), followed by chloroplast (150 proteins), cytoplasm (32 proteins), mitochondria (16 proteins), extracellular regions (6 proteins), and a small number localized to the plasma membrane (1 protein).

Table 5.10: Physicochemical Properties of R570

Dataset	Length (aa)	MW (Da)	pI
<i>A. thaliana</i> + R570	107–704	11,313.69–73,607.54	4.42–12
<i>O. sativa</i> + R570	107–704	11,313.69–73,607.54	4.28–12

Table 5.11: Subcellular Localization of R570

Dataset	Nuclear	Chloroplast	Cytoplasm	Mitochondria	Extracellular	Plasma Membrane
<i>A.thaliana</i> +R570	977	135	26	16	6	—
<i>O. sativa</i> + R570	978	150	32	16	5	1

5.7.3 *S. spontaneum*

The physicochemical properties of ERF proteins identified from the combined dataset of *S. spontaneum* and *Arabidopsis thaliana* showed considerable variation in protein length, molecular weight, and isoelectric point. The protein length ranged from 58 amino acids (*AT1G03800.1*) to 704 amino acids (*Sspon.02G0030540-1D*). The molecular weight varied between 6,716.48 Da and 73,607.54 Da, corresponding to the same minimum and maximum length proteins, respectively.

The isoelectric point (pI) ranged from 4.42 (*Sspon.01G0004940-1A*) to 12, with multiple proteins exhibiting the highest pI values. Subcellular localization prediction indicated that the majority of proteins were localized in the nucleus (310 proteins), followed by chloroplast (41 proteins), cytoplasm (14 proteins), mitochondria (3 proteins), and extracellular regions (1 protein).

Similarly, in the combined dataset of *S. spontaneum* and *Oryza sativa*, ERF proteins exhibited a wide range of physicochemical properties. The protein length ranged from 56 amino acids (*LOC_Os03g08460.1*) to 704 amino acids (*Sspon.02G0030540-1D*). The molecular weight ranged from 6,290.31 Da to 73,607.54 Da.

The isoelectric point (pI) ranged from 4.28 (*LOC_Os05g27930.2*) to 12, indicating the presence of both acidic and basic proteins. Subcellular localization analysis showed that most proteins were predicted to be localized in the nucleus (311 proteins), followed by chloroplast (56 proteins), cytoplasm (20 proteins), mitochondria (3 proteins), and extracellular regions (2 proteins).

Table 5.12: Physicochemical Properties of *S. spontaneum*

Dataset	Length (aa)	MW (Da)	pI
<i>A. thaliana</i> + <i>S. spontaneum</i>	99-5333	11141.46-595290.9	4.41-12
<i>O. sativa</i> + <i>S. spontaneum</i>	99-5333	11141.46-595290.9	4.28-12

Table 5.13: Subcellular Localization of *S. spontaneum*

Dataset	Nuclear	Chloroplast	Cytoplasm	Mitochondria	Extracellular	Plasma Membrane
<i>A.thaliana</i> + <i>S. spontaneum</i>	274	26	5	9	1	—
<i>O. sativa</i> + <i>S. spontaneum</i>	275	42	11	9	2	1

5.8. Genomic Synteny and Collinearity Analysis of the AP2/ERF Genes

5.8.1. Evolutionary Mechanisms of Gene Expansion

To understand the genomic organization and evolutionary history of the AP2/ERF genes identified in the previous sections, a comprehensive collinearity analysis was performed using MCScanX. Synteny—the conservation of gene order on chromosomes across different species—provides a "genetic map" that traces how these transcription factors were inherited and multiplied throughout the evolution of the *Saccharum* genus. By treating gene loci as nodes and their orthologous relationships as edges, we identified that the expansion of the sugarcane AP2/ERF superfamily was not a random occurrence but a structured result of large-scale duplication events.

5.8.2. Intra-species Collinearity and Segmental Duplication in *S. officinarum*

Analysis of the *S. officinarum* genome revealed that segmental duplications and Whole-Genome Duplication (WGD) were the primary drivers of gene proliferation. High-confidence syntenic blocks, such as Block 5-328 and Block 6-143, were identified across

homeologous chromosomes.

For instance, AP2/ERF members located on Chromosome 1A were found to have nearly identical "twins" on Chromosome 1B (e.g., *Soffic.01G0042210-1A.t1* paired with *Soffic.01G0040710-1B.t1*). The extremely low E-values (ranging from 3e-117 to 0.0) indicate that these gene pairs have remained virtually unchanged since the duplication of the sub-genomes, allowing the noble sugarcane to maintain a vast and redundant regulatory toolkit for environmental adaptation.

5.8.3. Interspecific Synteny: Tracing Inheritance in the R570 Hybrid

The relationship between the noble parent and the R570 hybrid was evaluated to determine how many AP2/ERF regulators were successfully passed down during breeding. The S.txt collinearity data showed a high level of "Syntenic Retention."

Major chromosomal regions, specifically Block 308 and Block 310, showed that AP2/ERF clusters on the *S. officinarum* 1A chromosome map directly to the R570 1Ag chromosome. This perfect collinearity (E-value = 0.0) confirms that the hybrid has faithfully inherited the core stress-response machinery—particularly the expansive Group VII (hypoxia) and Group IX (defense) clusters—from its noble ancestor.

5.8.4. Evolutionary Baseline from the Wild Ancestor (*S. spontaneum*)

Finally, by comparing the noble sugarcane to its wild ancestor, *S. spontaneum*, we traced the "one-to-many" expansion of the superfamily. A single ancestral locus in the wild species, such as *Sspon.01G0028640-1A* (Block 500-1), often corresponds to multiple duplicated copies in the *S. officinarum* genome.

This suggests that while the wild ancestor provided the foundational "blueprint" for AP2/ERF function, the noble species multiplied these templates to support its larger biomass and specialized physiology. Interestingly, the Soloist (Sor) genes remained in small, isolated networks, indicating that while other groups expanded for defense, these rare regulators were conserved as low-copy "specialists" throughout sugarcane evolution.

Table 5.14: Key Collinearity Blocks and Representative Gene Pairs

Block ID	Query Gene (Noble)	Target Gene (Reference)	E-value	Evolutionary Significance
5-328	Soffic.01G0042210-1A	Soffic.01G0040710-1B	3e-117	Internal duplication in Noble
308-577	Soffic.01G0039430-1A	R570.01Ag396100.1	0.0	Direct inheritance in Hybrid
500-1	Soffic.01G0008550-1A	Sspon.01G0028640-1A	4e-168	Expansion from Wild ancestor
310-249	Soffic.01G0027260-1A	R570.01Ag263900.1	4e-130	Conserved stress cluster

6. DISCUSSION

6.1. Genomic Proliferation and the "Nobilization" of the AP2/ERF Superfamily

6.1.1. The Meaning of the Results

The initial identification phase revealed a dramatic disparity in the AP2/ERF superfamily size across the *Saccharum* complex, ranging from 176 genes in the wild *S. spontaneum* to 1,336 genes in the noble *S. officinarum*. This 7.5-fold increase is a genomic signature of the complex polyploidization history of sugarcane. The data suggests that as sugarcane moved from its wild, ancestral state toward high-sugar "noble" varieties, the AP2/ERF regulatory network underwent a massive expansion. In the commercial hybrid R570, the identification of 1,020 genes represents a "genomic compromise," reflecting the mosaic nature of its sub-genomes inherited from both parents.

6.1.2. Comparison with Previous Studies

The gene counts identified in this study are consistent with the landmark genome assembly of *S. officinarum* by Zhang et al. (2018) and the R570 hybrid analysis by Garsmeur et al. (2018), which highlighted that sugarcane genomes contain multiple homeologous copies of almost every ancestral gene. While model plants like *Arabidopsis thaliana* (139 genes) and *Oryza sativa* (163 genes) possess relatively compact AP2/ERF families (Nakano et al., 2006), the expansion seen in *Saccharum* is among the largest recorded in the plant kingdom. This follows the "dosage effect" theory proposed by Birchler and Veitia (2012), where polyploid plants maintain high gene copy numbers to increase the total output of critical regulatory proteins.

6.1.3. Biological Significance

The expansion of the AP2/ERF family has profound implications for environmental resilience. AP2/ERF transcription factors are the primary "first responders" to abiotic stresses (Riechmann and Meyerowitz, 1998). By maintaining over 1,300 copies of these regulators, *S. officinarum* effectively creates a "genetic safety net." This redundancy ensures that if one copy of a stress-response gene is silenced, several other homeologous copies remain functional to protect the plant's high sugar content.

6.2. Structural Conservation vs. Functional Diversification

6.2.1. The Meaning of the Results

Despite the vast difference in gene numbers, the domain analysis revealed a striking level of structural conservation. The majority of the proteins exhibited the characteristic single AP2 DNA-binding domain (PF00847). The consistently low E-values ($1e^{-25}$ to $1e^{-34}$) indicate that the core "machinery" that allows these proteins to bind to DNA has remained intact despite millions of years of evolution.

6.2.2. Comparison with Previous Studies

The stability of the AP2 domain architecture observed here aligns with the findings of Rashid et al. (2012) in *Oryza sativa*. These studies established that, while the number of genes in the ERF subfamilies can vary widely across species, the amino acid sequence within the AP2 domain itself is under strong purifying selection. Our results confirm that sugarcane follows this universal plant blueprint, as the domain profiles matched the reference benchmarks established by Marchler-Bauer et al. (2017).

6.2.3. Biological Significance

The high conservation of the AP2 domain suggests that sugarcane has prioritized "Functional Stability." By keeping the DNA-binding domain unchanged while duplicating the genes, the plant can evolve new ways to regulate the genes without breaking the fundamental ability of the protein to function. This structural "rigidity" combined with genomic "flexibility" is likely what allowed sugarcane to adapt to diverse tropical climates while maintaining its core physiological processes.

6.3. Architecture of Conserved Motifs: Structural Blueprints and Functional Innovation

6.3.1. The Meaning of the Results

The MEME suite analysis revealed a highly organized architectural framework within the sugarcane AP2/ERF superfamily. The identification of Motifs 1, 2, and 3 as the "core" elements across all three lineages—*S. officinarum*, *S. spontaneum*, and the R570 hybrid—confirms that these represent the essential amino acid sequences of the AP2 DNA-binding domain. The consistent positioning of these core motifs, even when compared to the dicot *Arabidopsis*, demonstrates that the fundamental biophysical mechanism for DNA recognition has remained "frozen" throughout evolutionary time.

However, the "auxiliary" Motifs 4–10 showed distinct, subgroup-specific distributions. This suggests that while the "engine" of the protein (the AP2 domain) is conserved, the "accessories" (the auxiliary motifs) have diversified to allow different AP2/ERF groups to perform specialized tasks, such as protein-protein interaction or nuclear localization.

6.3.2. Comparison with Previous Studies

The presence of three core motifs forming the AP2 domain is in total agreement with the structural models established by Allen et al. (1998) and later confirmed in monocots by Rashid et al. (2012). Our findings that *S. officinarum* and R570 exhibit "repetitive motif patterns" (multiple genes with identical motif layouts) is a classic molecular signature of polyploid expansion.

As noted by Vanneste et al. (2014), following a whole-genome duplication, duplicated genes (paralogs) often maintain identical motif architectures for millions of years. This explains why we see "one-to-many" orthologous relationships where several sugarcane genes perfectly mirror a single *Oryza sativa* gene. In contrast, the "streamlined" and "compact" motif organization found in *S. spontaneum* reflects its status as an ancestral representative, possessing the original "blueprint" before the massive duplication events of the noble lineages (Zhang et al., 2018).

6.3.3. Biological Significance

The high degree of motif collinearity—where the order of motifs 1 through 10 stays the same between *Oryza sativa* and sugarcane—is biologically critical. It suggests that these transcription factors are under intense purifying selection. If the order of these motifs were to change, the protein would likely lose its ability to fold correctly or interact with its target DNA sequences.

The retention of these structural motifs in the R570 hybrid is particularly significant for sugarcane breeding. It proves that despite the "genomic shock" of hybridization, the functional potential of the AP2/ERF family was not compromised. The auxiliary motifs (4–10) likely harbor the specific "activation" or "repression" signals that allow sugarcane to fine-tune its growth and stress responses, providing the regulatory complexity needed for the high biomass production seen in modern cultivars.

6.3.4. Limitations of the Study

While the MEME and TBtools analysis provides a clear visual map of protein conservation, it is limited to sequence-based motifs. It does not account for the three-dimensional (3D) folding of the proteins. Two proteins might have different auxiliary motifs but still fold into a similar shape that performs the same function. Furthermore, the maximum setting of 10 motifs might overlook rare, lineage-specific motifs that only appear in a few specialized *Saccharum* genes. Future studies employing AlphaFold2 protein modeling would be beneficial to determine how these 10 motifs interact in a 3D space to facilitate DNA binding.

6.4. Phylogenetic Diversification and Subgroup-Specific Expansion

6.4.1. The Meaning of the Results

The phylogenetic reconstruction of the AP2/ERF superfamily across three *Saccharum* lineages reveals a highly structured evolutionary landscape. By utilizing both *Arabidopsis* (dicot) and *Oryza sativa* (monocot) as reference frameworks, we successfully partitioned the massive sugarcane datasets into established subgroups (I–X and I–XIV).

The most significant finding is the uneven distribution of genes across these clades. While the ancestral *S. spontaneum* possesses a compact version of every major group, the noble *S. officinarum* and the R570 hybrid exhibit explosive expansion in Groups VII, III, and IX. The presence of over 250 genes in Group VII of *S. officinarum* indicates that certain regulatory "chapters" of the sugarcane genome have been copied much more frequently than others. Furthermore, the identification of Soloist (SOR) genes and a significant proportion of Unclassified (UN) members (up to 182 in *S. officinarum*) suggests that sugarcane has developed unique genetic innovations that no longer match the standard models of *Arabidopsis* or *Oryza sativa*.

6.4.2. Comparison with Previous Studies

The classification of sugarcane ERF members into Groups I through X aligns with the foundational nomenclature established by Nakano et al. (2006). However, the expansion of Group VII (associated with waterlogging and hypoxia) and Group IX (associated with biotic defense) is much more pronounced in sugarcane than in other tropical grasses.

Our results mirror the findings of Rashid et al. (2012) in *Oryza sativa*, particularly in the consistent absence of Groups XI, XII, and XIV in the *Saccharum* lineage when using the *Oryza sativa*-reference

tree. This suggests that these specific subgroups may have been lost or significantly diverged within the *Poaceae* family prior to the emergence of the *Saccharum* genus. The resolution of lineage-associated clusters like VI-L and Xb-L further supports the "Lineage-Specific Expansion" (LSE) model proposed by Lespinet et al. (2002), where certain gene families multiply rapidly to meet the specific environmental needs of a species.

6.4.3. Biological Significance

The dominance of Groups VII, III, and IX carries heavy biological weight. Members of Group VII are known as the "master regulators" of flooding tolerance (the N-end rule pathway), while Group III (DREB) genes are critical for drought and cold acclimation (Sakuma et al., 2002).

The fact that *S. officinarum* and R570 have prioritized the expansion of these specific groups suggests that the "nobilization" process selected for a highly robust stress-response system. This "genomic buffering"—where hundreds of genes are available to handle a single type of stress—likely explains why sugarcane can maintain high biomass and sugar accumulation even under the fluctuating environmental conditions of the tropics. The Soloist (SOR) genes, although few in number, are highly conserved, pointing to their role as essential "specialist" regulators that the plant cannot afford to lose or mutate.

6.4.4. Limitations of the Study

A primary limitation of the phylogenetic analysis is the "Long-Branch Attraction" artifact that can occur in Maximum Likelihood trees with such large datasets (1,300+ sequences). This may explain why a large number of genes (UN) remained unclassified; they may have diverged so significantly that they no longer share enough sequence similarity to cluster with the *Arabidopsis* or *Oryza sativa* benchmarks. Additionally, while the tree shows evolutionary relationships, it does not prove functional identity. Just because a sugarcane gene clusters with an *Arabidopsis* Group VII gene does not guarantee it performs the exact same role in the cell. Future expression profiling (RT-qPCR) is necessary to see if these expanded groups are actually active during stress.

6.5. Structural Architecture of AP2/ERF Genes

6.5.1. The Meaning of the Results

The analysis of exon–intron organization across *S. officinarum*, *S. spontaneum*, and the R570 hybrid reveals a predominantly intronless or intron-poor structural

framework. In all three lineages, the majority of AP2/ERF genes consist of a single, continuous coding sequence (CDS) block. This "compact" gene architecture is a defining characteristic of the ERF subfamily. However, the presence of a smaller subset of genes with multiple exons (interrupted by black intronic lines) suggests that while the superfamily is generally simple, certain clades have maintained complex splicing requirements. The variation observed in the length of the yellow UTR regions further suggests that sugarcane has developed diverse regulatory "handles" at the beginning and end of these genes to control how quickly they are translated into proteins.

6.5.2. Comparison with Previous Studies

These findings are in high concordance with the foundational "Intronless Model" of ERF genes established by Nakano et al. (2006) in *Arabidopsis* and Rashid et al. (2012) in *Oryza sativa*. The stability of these simple structures across the wild ancestor (*S. spontaneum*), the noble species (*S. officinarum*), and the hybrid (R570) indicates that the AP2/ERF family is under intense structural constraints. According to the "Speedy Transcription" theory proposed by Jeffares et al. (2008), genes that lack introns are evolutionarily favored for environmental stress responses because they can be transcribed and exported from the nucleus much faster than multi-exon genes, allowing for a near-instantaneous cellular reaction to drought or heat.

6.5.3. Biological Significance

The transition from the "streamlined" structures in the wild ancestor to the "expanded but conserved" structures in the cultivated varieties is biologically critical. Because most sugarcane AP2/ERF genes lack introns, they are less likely to undergo "Alternative Splicing." This means the plant produces a very predictable and stable version of the protein every time the gene is turned on. For a high-biomass crop like sugarcane, having a reliable and fast-acting regulatory toolkit is essential for maintaining growth under fluctuating tropical conditions. The complex, multi-exon genes identified in this study likely represent the AP2 and RAV subfamilies, which involve more intricate developmental roles beyond simple stress signaling.

6.5.4. Limitations of the Study

A primary limitation of this gene structure analysis is that it relies on automated genome annotations. In complex polyploids like sugarcane, sequencing gaps can sometimes lead to the "false" identification of an intronless gene when, in reality, the intron was simply not captured in the assembly. Furthermore, while TBtools provides an excellent visual map of the CDS and

UTR, it does not show Alternative Polyadenylation (APA) sites within the UTRs. Future studies using Full-length Isoform Sequencing (Iso-Seq) would be necessary to verify if any of these "simple" genes actually produce multiple versions of mRNA that our current genomic map cannot see.

6.6. Chromosomal Localization and Genomic Distribution

6.6.1. The Meaning of the Results

The chromosomal mapping across *S. officinarum*, *S. spontaneum*, and the R570 hybrid provides a high-resolution view of how the AP2/ERF superfamily is physically anchored within the complex sugarcane genome. The results demonstrate that these genes are not restricted to specific "stress-response" chromosomes but are instead widely dispersed across all basic chromosome groups (Chr01–Chr10 in the noble and hybrid lines; Chr1–Chr8 in the wild ancestor).

A key observation is the uneven gene density and the presence of localized clusters. While the genes are spread out, certain chromosomal regions act as "hotspots" with high concentrations of AP2/ERF loci. This suggests that the expansion of this family occurred through a combination of whole-genome duplication (providing the broad distribution) and tandem duplications (creating the localized clusters).

6.6.2. Comparison with Previous Studies

The distribution of AP2/ERF genes across multiple homologous sets (such as A–H in *S. officinarum* and A–F in R570) serves as a direct molecular validation of the polyploid nature of the *Saccharum* genus, as documented by Zhang et al. (2018) and Garsmeur et al. (2018). In simpler diploid relatives like *Sorghum bicolor*, AP2/ERF genes also show a dispersed distribution, but the sheer scale of the 1,336 genes identified in *S. officinarum* represents one of the most complex regulatory maps in the Poaceae family. The clustering observed in our results is consistent with the findings of Liu et al. (2013), who noted that tandemly duplicated genes in plants are often involved in rapid adaptation to abiotic stressors, such as those faced by sugarcane in tropical environments.

6.6.3. Biological Significance

The wide dispersal of AP2/ERF genes across the entire genome is a critical evolutionary strategy for genomic buffering. By having these vital transcription factors located on every

chromosome and every homologous copy, the plant ensures that the loss or silencing of a single chromosomal segment (due to environmental damage or mutation) does not lead to a total failure of the stress-response system.

Furthermore, the gene-rich clusters identified on specific chromosomes likely represent evolutionary "innovation zones." These clusters allow for the co-regulation of multiple AP2/ERF members, enabling the plant to mount a coordinated and massive transcriptional response to environmental stimuli. This architectural complexity is likely what supports the high biomass and exceptional adaptability of modern sugarcane cultivars.

6.6.4. Limitations of the Study

While the mapping performed using TBtools provides an accurate physical location of the genes, it is based on the linear genomic sequence. It does not account for the three-dimensional (3D) chromatin architecture. In reality, genes located on different chromosomes may physically interact in the nucleus to regulate growth. Additionally, because the sugarcane genome is still being refined, some genes mapped to "unanchored scaffolds" might have been excluded from the primary chromosome visualization. Future research using Hi-C sequencing (High-throughput Chromosome Conformation Capture) would be beneficial to see how these dispersed AP2/ERF genes cluster together in 3D space to function as a unified regulatory network.

6.7. Physicochemical Diversity and Subcellular Compartmentalization

6.7.1. The Meaning of the Results

The analysis of physicochemical properties across the *Saccharum* complex, *Arabidopsis*, and *Oryza sativa* reveals a high degree of structural heterogeneity within the AP2/ERF superfamily. The significant ranges in protein length (56 to 1,780 amino acids) and molecular weight (6.29 kDa to 199.34 kDa) suggest that while these proteins share a conserved DNA-binding domain, their flanking regions have diversified extensively.

The isoelectric point (pI) values, spanning from acidic (4.1) to highly basic (12.0), indicate that the superfamily contains proteins capable of functioning under diverse cellular pH conditions and interacting with various co-factors. Most importantly, the consistent prediction of nuclear localization for the vast majority of proteins (e.g., 1,234 in *S. officinarum*) confirms their primary biological identity as transcription factors that must enter the nucleus to regulate gene expression.

6.7.2. Comparison with Previous Studies

The variation in physicochemical attributes observed here mirrors the findings of Lata et al. (2014) in foxtail millet and Rashid et al. (2012) in *Oryza sativa*, where ERF proteins were found to be highly diverse in size despite domain conservation. The extreme length observed in some *S. officinarum* proteins (1,780 aa) is notably larger than typical *Arabidopsis* ERFs, which may be attributed to the complex polyploid nature of sugarcane where some gene models may have fused or expanded through extensive duplication (Zhang et al., 2018).

The finding that a small percentage of proteins are predicted to localize in the chloroplast or mitochondria aligns with the "Dual-Targeting" hypothesis proposed by Krupinska et al. (2013). This suggests that certain AP2/ERF members may participate in retrograde signaling, where they sense stress in organelles and communicate that information back to the nucleus.

6.7.3. Biological Significance

The dominance of the nucleus as the primary site of localization is biologically critical for transcription factors. It confirms that the identified sequences possess functional Nuclear Localization Signals (NLS) that allow them to bind to specific promoter elements like the DRE/CRT or GCC-box.

Furthermore, the diversity in isoelectric points is significant for sugarcane's adaptability. Proteins with highly basic pI values (close to 12.0) are often more effective at binding to negatively charged DNA under stress conditions. The presence of proteins in the chloroplast (up to 170 in *S. officinarum*) is particularly relevant for sugarcane, as these proteins may be involved in protecting the photosynthetic machinery—the source of its high sugar content—from photo-oxidative stress and high temperatures.

6.7.4. Limitations of the Study

A limitation of this analysis is the reliance on in silico prediction tools (such as WoLF PSORT or ProtParam). While these algorithms are highly accurate for model species, their performance on polyploid sugarcane proteins has not been fully validated with "wet-lab" experiments. Factors such as post-translational modifications (e.g., phosphorylation or sumoylation) can drastically change a protein's actual pI and its localization in a living cell. Future research using Green Fluorescent Protein (GFP) tagging in sugarcane protoplasts is necessary to confirm that the proteins predicted to be in the chloroplast or mitochondria actually function within those organelles during environmental stress.

6.8. Genomic Synteny and Collinearity Analysis

6.8.1. The Meaning of the Results

The synteny analysis using MCScanX serves as a "genomic bridge," connecting the physical location of AP2/ERF genes to their evolutionary history across the *Saccharum* complex. By identifying highly conserved gene orders (collinearity), we have demonstrated that the expansion of this superfamily was a structured process rather than a series of random mutations.

The identification of key blocks (e.g., Block 5-328 and Block 500-1) provides evidence that the 7.5-fold increase in gene count in the noble sugarcane was driven primarily by Segmental Duplications and Whole-Genome Duplications (WGD). This means that instead of single genes duplicating, entire chromosomal segments containing multiple AP2/ERF regulators were copied simultaneously, creating a robust and redundant regulatory network.

6.8.2. Comparison with Previous Studies

The "Syntenic Retention" observed between *S. officinarum* and the R570 hybrid (with E-values of 0.0) is in full agreement with the genome assembly findings of Garsmeur et al. (2018). These results confirm that modern sugarcane hybrids are a high-fidelity "mosaic" of their noble and wild parents.

Furthermore, the "one-to-many" mapping from *S. spontaneum* to *S. officinarum* mirrors the "dosage effect" models proposed by Birchler and Veitia (2012). Our data shows that while the wild ancestor provided the original structural template, the "nobilization" process identified by Zhang et al. (2018) resulted in the massive multiplication of these templates to support the specialized physiological needs—such as high sugar accumulation and rapid growth—of cultivated sugarcane.

6.8.3. Biological Significance

The evolutionary strategy of sugarcane is clearly revealed through this collinearity. By maintaining "nearly identical twins" of AP2/ERF members on homeologous chromosomes (e.g., Chromosomes 1A and 1B), *S. officinarum* has developed a form of genetic insurance. If one copy of a critical regulator is compromised, the high-confidence syntenic partner (the twin) can maintain the necessary stress-response function.

The high level of inheritance found in the R570 hybrid (Block 308 and 310) is particularly important for agriculture. It proves that the "core stress-response machinery," especially the

expansive Group VII (hypoxia) and Group IX (defense) clusters, is successfully passed down during breeding. In contrast, the Soloist (Sor) genes remained in small, isolated networks, suggesting they are "dosage-sensitive" regulators that the plant must keep in low copies to avoid disrupting delicate developmental balances.

6.8.4. Limitations of the Study

While the synteny analysis successfully traces the "pathways" of gene inheritance, it is limited to sequence-based collinearity. It does not account for epigenetic changes (such as DNA methylation) that may silence one of the duplicated copies despite it being physically present in a syntenic block. A gene pair may have an E-value of 0.0 (perfect sequence match), but if one copy is silenced by the plant's internal cellular machinery, its functional contribution is lost. Future research using RNA-Seq is required to confirm if all duplicated syntenic copies are equally "active" in the sugarcane plant during environmental stress.

7. SUMMARY

The *Saccharum* complex, comprising high-sugar "noble" varieties, wild ancestors, and complex commercial hybrids, represents one of the most challenging yet vital genomic frontiers in tropical agriculture. Central to the environmental resilience and high biomass of sugarcane is the AP2/ERF transcription factor superfamily, which serves as the primary regulatory "first responder" to abiotic and biotic stresses. This study provides a comprehensive, genome-wide audit of the AP2/ERF superfamily across three key lineages: *S. officinarum* (noble), *S. spontaneum* (wild), and the R570 hybrid. By employing advanced bioinformatic tools and comparative synteny analysis, this research decodes the structural and evolutionary mechanisms that allowed this gene family to expand into one of the largest regulatory networks recorded in the plant kingdom.

The investigation revealed a dramatic 7.5-fold increase in the superfamily size, ranging from 176 genes in the wild ancestor to 1,336 genes in the noble sugarcane, with the R570 hybrid maintaining a "genomic compromise" of 1,020 genes. Despite this massive proliferation, the study identified a striking level of structural conservation, with core AP2 DNA-binding domains and motif architectures (Motifs 1–3) remaining largely "frozen" across millions of years. Evolutionary analysis, including collinearity mapping and Ka/Ks ratios, demonstrates that this expansion was not random but driven by Segmental and Whole-Genome Duplications (WGD). This "dosage effect" has created a robust "genetic safety net," where redundant homeologous gene copies (or "twins") ensure functional stability and genomic buffering against environmental fluctuations.

A significant outcome of this research is the identification of lineage-specific expansions in Groups VII (hypoxia), III (drought), and IX (defense), highlighting the "nobilization" process's focus on robust stress-response systems. While the majority of genes exhibited an intronless, "speedy transcription" architecture favored for rapid stress signaling, a subset of complex multi-exon genes was identified as potential specialists. Furthermore, synteny analysis confirmed the high-fidelity inheritance of core stress clusters from noble ancestors to modern hybrids. This study establishes a definitive structural and functional framework for the sugarcane AP2/ERF superfamily, providing the molecular targets necessary for Marker-Assisted Selection (MAS) and Genome Editing (CRISPR) to enhance climate resilience and sugar yield in future sugarcane cultivars.

8. CONCLUSION AND FUTURE SCOPE

This research successfully established a high-resolution genomic and evolutionary baseline for the AP2/ERF transcription factor superfamily across the *Saccharum* complex, moving beyond simple identification to a multi-dimensional analysis of *S. officinarum*, *S. spontaneum*, and the R570 hybrid. The major findings revealed a dramatic 7.5-fold expansion of the superfamily—from 176 genes in the wild ancestor to 1,336 in the noble variety—reflecting a massive "nobilization" of the regulatory network. This expansion, driven primarily by Segmental and Whole-Genome Duplications (WGD), aligns with the "dosage effect" theory proposed by Birchler and Veitia (2012), where polyploid plants maintain high gene copy numbers to provide a "genomic safety net" against environmental stress. The study confirmed that while the superfamily expanded, the core AP2 DNA-binding domain remained under intense purifying selection, maintaining structural stability while allowing for functional innovation through auxiliary motifs. By identifying lineage-specific expansions in Groups VII (hypoxia), III (drought), and IX (defense), this work provides a definitive molecular framework for transitioning sugarcane breeding from traditional phenotypic selection to a proactive, genomically-informed strategy.

The practical applications of this genomic library are immediate, providing a foundation for Precision Molecular Breeding and Marker-Assisted Selection (MAS) to target the robust stress-response clusters identified in this study. Looking ahead, the future scope of this work is significant, particularly in the integration of 3D Protein Modeling (AlphaFold2) to understand how specific motifs facilitate DNA recognition in a polyploid environment. There is a clear opportunity to implement Machine Learning (ML) models to analyze these 1,300+ genes alongside climatic data to predict cultivar performance under shifting environmental pressures. Furthermore, the identification of "master regulators" in Group VII and "Soloist" clades offers high-priority targets for CRISPR/Cas9 genome editing to enhance waterlogging and drought resilience. By combining these computational tools with Functional Transcriptomics (RNA-Seq), future research can pinpoint which redundant homeologs are truly active during stress, ensuring the long-term biological security and climate-resilience of the global sugarcane industry.

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